Strathcona Hydrogenated Renewable Diesel (SHRED)

JDB 2.04.1 Deep Foundation Philosophy

S1RD-42-CAL-215-003-01

P.ENG STAMP	PERMIT TO PRACTICE STAMP
PROFESSIONAL ENGINEER ALBERTA CHRIS PALANQUE 172427 16 Nov. 2022	**PERMIT TO PRACTICE** **FLUOR CANADA LTD** RM SIGNATURE: RM APEGA ID #: 66911 DATE: 2022-11-17 **PERMIT NUMBER: P000778** The Association of Professional Engineers and Geoscientists of Alberta (APEGA)

Checking Level: 3B

DOC REV	REVISION DESCRIPTION / STATUS	DATE ISSUED	FLUOR ORIGINATOR	CHECKER FLUOR ENGINEERING CHECKER	APPROVAL FLUOR DISCIPLINE LEAD
0	ISSUED FOR DESIGN	16-NOV-22	C. PALANQUE	C. WALSH	B. KAPCSOS
			Digitally signed by pal59246 DN: cn=pal59246, email=Chris.Palanque@fluor.com Date: 2022.11.16 15:32:47 -07'00'	Colton Walsh Digitally signed by wal10531 Date: 2022.11.16 15:36:53 -07'00'	Digitally signed by kap51978 Date: 2022.11.16 16:32:59 -07'00'

Risk Notice

Does this calculation have any assumptions or risk items that need to be followed up with? Yes ☐ No ☒

If yes, anticipated date information will be available: N/A

Followed up by: Date: Signature:

Revision History

Please be descriptive on revisions to original calculation.

Rev No.	Originator (Name)	Checker (Name)	Revision Date	Revision Description
0	Chris Palanque	Coltin Walsh	16-Nov-2022	Issued for Design

1 RISK AND UNCERTAINTY LOG

The purpose of the table outlines the uncertainities, mitigation strategy, remaining risk at the time of design. Assumptions to be re-validated once the appropriate information is made available.

Seq # (PDR # If avail)	Uncertainty	Mitigation	Remaining Risk	Resolution (include approx date)
	None			

2 COMPUTER FILE INDEX

Title: JDB 2.04.1 Deep Foundation Philosophy

Calculation Number: S1RD-42-CAL-215-003-01

The purpose of this form is to keep track of calculation computer files and to assign filenames using a standard format. This will assist in organizing data and in retrieving data at a later date.

List of native file calculations:

FILENAME	DESCRIPTION	DIRECTORY LOCATION
EP Steel Pipe Pile Axial Capacities.xlsx	Pie compression & tension resistances	215_STRUCTURAL\215E_ENGINEERING_DESIGN_DATA\215E_014_R_Proj_Calculations_Sketches\S1RD-42-CAL-215-003-01 - JDB 2.04.1 Deep Fdn Philosophy
Several files (.lp11d)	LPile analyses for free and fixed piles	215_STRUCTURAL\215E_ENGINEERING_DESIGN_DATA\215E_014_R_Proj_Calculations_Sketches\S1RD-42-CAL-215-003-01 - JDB 2.04.1 Deep Fdn Philosophy\LPile runs

3 GENERAL DESCRIPTION

This document summarizes the deep foundation recommendations from the Geotechnical EOR (Tetra Tech) to be followed by the Fluor structural team for the RDU Plant and immediately adjacent Storage Tank Farm during the EP phase of the Strathcona Hydrogenated Renewable Diesel (SHRED) Project.

This document does not include considerations for the structural resistance of the deep foundations (except during driving) which are the responsibility of the Structural EOR.

4 BASIS OF DESIGN

4.1 Detailed Design Geotechnical Report

Tetra Tech's Geotechnical Report entitled "Geotechnical Evaluation for Detailed Design – Strathcona Refinery SHRED Project Edmonton, Alberta", reference ENG.EGEO04009-02.002, Rev 0 dated 09-Sep-2022.

4.2 Checking Level

The following checking level applies to this calculation. Descriptions for the checking levels are found in the Activity Plan – Civil Structural Engineering.

Check where applicable	Checking Level	Description
	1B	Detailed check in accordance with governing work instruction or practice
X	2B	Selective Check with particular attention to critical and interface points.
	3B	Spot check, applied as established by project lead
	4B	Self-Check only

5 JDB 2.04.1 DEEP FOUNDATION PHILOSOPHY

See attached in subsequent pages.

DEEP FOUNDATION PHILOSOPHY

1	19 Sep 2022	Issued for EP Phase	Brett Kapcsos	Jason Dias
0	10 Feb 2022	Issued for FEED Phase	Brett Kapcsos	Jason Dias
Rev.	**Date**	**Revision Description**	**Lead Engineer**	**Lead Designer**

DEEP FOUNDATION PHILOSOPHY

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DEEP FOUNDATION PHILOSOPHY

1.0 INTRODUCTION

This document summarizes the deep foundation recommendations from the Geotechnical EOR (Tetra Tech) to be followed by the Fluor structural team for the RDU Plant and immediately adjacent Storage Tank Farm during the EP phase of the Strathcona Hydrogenated Renewable Diesel (SHRED) Project.

This document does not include considerations for the structural resistance of the deep foundations (except during driving) which are the responsibility of the structural engineer.

2.0 REFERENCE DOCUMENTS

- Tetra Tech's Geotechnical Report entitled "Geotechnical Evaluation for Detailed Design – Strathcona Refinery SHRED Project Edmonton, Alberta", reference ENG.EGEO04009-02.002, Rev 0 dated 09-Sep-2022.

- Tara, D. and Verbeek, G. 2016. Development of a Screening Tool for Impact Hammer Selection for Installation, Testing and Damage Mitigation of Steel Pipe and H-Piles, GeoVancouver.

3.0 GENERAL

3.1. Ground Conditions

The general geological sequence of the site generally consists of shallow fill materials over native clays over glacial clay till over sands / gravels (not always present) over weathered bedrock (clay shale and sandstone). Suitable geological materials for founding piles are reasonably shallow with glacial tills generally encountered at 4m to 5m depth below existing grade, and weathered bedrock at typically 12m to 19m depth.

A general bathtub excavation from the originally existing grade (i.e. approx. Plant EL 2,170 ft) down to Plant EL 2,164 ft is under construction in the RDU unit and in the Tank Storage Farm to remove all existing buried foundations and utilities as well as some soil contamination. An additional central excavation below the bathtub down to EL 2,157'- 6" will also be carried out in the Tank Storage Farm to remove additional contamination. These excavations will be backfilled to the rough grading elevations using select engineered fill compacted to min. 98% SPMDD. Refer to the AFC Civil Early Works excavation drawings S1RD-RD-DTL-210-900-01 & -02 and the rough grading drawings S1RD-RD-DTL-210-901-01 & -02 for further details.

3.2. Groundwater

During their investigation in July 2021, Tetra Tech found groundwater at a depth of only 1.6m below existing grade in one borehole but more generally at minimum depth of 3.4m. The existing grade is not flat and, based on the records reported by Tetra Tech, groundwater should be expected to be encountered between EL 657.5m (2,157 ft) and EL 659.5m (2,164 ft). However, groundwater level could be seasonal and may therefore be encountered at a slightly higher elevation in the Spring and/or in the Fall.

DEEP FOUNDATION PHILOSOPHY

3.3. Seismic Class

Per Tetra Tech's Geotechnical Report section 6.6, the seismic classification for the site may be taken as Site Class D.

3.4. Pile Types

The pile types identified as being suitable for the site ground conditions by Tetra Tech include:

- Cast-In Place (CIP) Piles
- Continuous Flight Auger (CFA) Piles
- Belled Piles
- Steel Pipe Piles
- Helical Piles.

Driven steel pipe piles and helical piles have been selected as the preferred pile types for the project (refer to Project Decision Record PDR-006). As such, this document only includes recommendations for steel pipe piles and helical piles.

4.0 DRIVEN STEEL PIPE PILES

4.1. Recommended Pile Sizes

The recommended pile outside diameters and wall thicknesses are as follows:

Outside Diameter (D)		API (t_min)(1)	Adopted Wall Thickness (t)		D/t	Remark
(inch)	(mm)	(mm)	(inch)	(mm)		
10 – ¾	273	9.1	3/8	9.5	28.7	*Satisfies API guideline*
12 – ¾	324	9.6	3/8	9.5	34.1	*Satisfies API guideline*
16	406	10.4	3/8	9.5	42.7	*Does not satisfy API guideline but meets D/t < 60*
24	610	12.5	1/2	12.7	48.0	*Satisfies API guideline*
30	762	14.0	5/8	15.9	47.9	*Satisfies API guideline*
36	914	15.5	3/4	19.0	48.1	*Satisfies API guideline*

Notes: 1. Min wall thickness, t_min = D/100 + 6.35mm for driving purposes per API guideline.
2. D/t should be less than 60 based on D. Tara & G. Verbeek (2016).

Tetra Tech conducted pile driveability (GRLWEAP) analyses using the above selected pile wall thickness and presented the results in Appendix E of the Geotechnical Report. The results of these preliminary analyses support the selected pile wall thicknesses and steel grade (i.e. ASTM A252 Grade 3 – fy = 310 MPa).

A total corrosion allowance of **1.5mm** on the pile wall thickness shall be considered for detailed design purposes based on the recommendations provided in the Tetra Tech's Geotechnical Report section 12.0. This corrosion allowance shall be applied to outer face of the pile and not to the inner one.

DEEP FOUNDATION PHILOSOPHY

4.2. Single Pile Ultimate Compression Resistance

Based on the recommendations from Tetra Tech, a single pile ultimate resistance in compression can be estimated from the following equation:

$$\mathbf{Q_u = Q_{bu} + Q_{su} = q_b \times A_b + q_s \times A_s}$$

where: Q_u = Ultimate geotechnical pile compression resistance (kN)
Q_{bu} = Ultimate end-bearing resistance (kN)
Q_{su} = Ultimate shaft resistance (kN)
q_b = ultimate end bearing resistance (kPa)
A_b = gross end-bearing area (m^2).
q_s = ultimate skin friction (kPa)
A_s = shaft area (m^2).

For the recommended ultimate skin friction and end bearing values, please refer to Tetra Tech's Geotechnical Report section 9.1.3. **Note the recommendation to neglect the skin friction within the select engineering fill placed during the Early Civil Works or within all soils above EL 2162ft, whichever is the most onerous, for the entire site**.

A geotechnical resistance factor (Φ) of 0.5 in compression is recommended based on the performed pile PDA tests (refer to Tetra Tech's Geotechnical Report section 9.1.3).

Ultimate and factored pile geotechnical resistance values in compression based on Tetra Tech recommendations are presented below for pile diameters ranging from 273mm to 914mm.

<u>Northern Area (see Tetra Tech's Geotechnical Report Fig 1 – also attached in Appendix):</u>

The recommended soil profiles and design parameters are as follows (refer to Tetra Tech's Geotechnical Report section 9.1.3):

Table 8A: Driven Steel Pile Shaft and End-Bearing Pile Parameters For Northern Area (Indicated on Figure 1)

Elevation [ft(m)]	Soil Type	Shaft Resistance			Base Resistance	
		Ultimate (kPa)	Factored for Compression (kPa)	Factored for Tension (kPa)	Ultimate (kPa)	Factored for Compression (kPa)
Above 2157 (Above 657.5)	Clay	30	15	10	-	-
2157 to 2145 (657.5 to 654.0)	Clay Till	45	22	15	-	-
2145 to 2141 (654.0 to 652.6)	Clay Till	60	30	21	800	400
2141 to 2109 (652.6 to 643.0)	Sand/Sand and Gravels	120	60	42	3,000*	1,500
Below 2109 (643.0)	Weathered Bedrock	120	60	42	3,000*	1,500

*Base resistance reduced considering bottom of pile may be partially plugged.

DEEP FOUNDATION PHILOSOPHY

The corresponding calculated pile ultimate and factored compression resistances are as tabulated below for the facilities located within the footprint of the Early Civil Works bathtub excavation down to EL. 2,164 ft. For the excavation limits of the bathtub down to EL. 2,164 ft, please refer to the AFC civil drawings S1RD-RD-DTL-210-900-01 and -02. For pile locations where the Early Works excavation is deeper than EL. 2,164 ft (i.e. part of the Storage Tank area), the depth over which the pile shaft friction is to be neglected needs to be increased accordingly

For the native calculation file, refer to EP Steel Pipe Pile Axial Capacities.xlsx located under: 215_STRUCTURAL\215E_ENGINEERING_DESIGN_DATA\215E_001_Job_Data_Book\Section 2 - Structural Engineering Instructions\JDB 2.04.1 Deep Foundation Philosophy\EP Pile Axial Capacities.

							Ultimate Shaft Resistance - Compression (kN)						Ultimate	Unfactored	**Factored**
Dia.	Finished Grade (El.)		Depth		Tip (El.)		Clay	Clay Till	Clay Till	Sand / Gravel	Bedrock	Total	End-bearing	Resistance	**Resistance**
(m)	(ft)	(m)	(ft)	(m)	(ft)	(m)	(kN)	(kN)	(kN)	(kN)	(kN)	(kN)	(kN)	(kN)	**(kN)**
0.273	2,169	661.1	40	12.2	2,129	648.9	39	141	63	376	0	620	176	795	**398**
0.273	2,169	661.1	45	13.7	2,124	647.4	39	141	63	533	0	776	176	952	**476**
0.273	2,169	661.1	50	15.2	2,119	645.9	39	141	63	690	0	933	176	1109	**554**
0.273	2,169	661.1	55	16.8	2,114	644.3	39	141	63	847	0	1090	176	1266	**633**
0.273	2,169	661.1	60	18.3	2,109	642.8	39	141	63	1004	0	1247	176	1423	**711**
0.273	2,169	661.1	65	19.8	2,104	641.3	39	141	63	1004	157	1404	176	1579	**790**
0.324	2,169	661.1	40	12.2	2,129	648.9	47	168	74	447	0	735	247	983	**491**
0.324	2,169	661.1	45	13.7	2,124	647.4	47	168	74	633	0	921	247	1169	**584**
0.324	2,169	661.1	50	15.2	2,119	645.9	47	168	74	819	0	1108	247	1355	**677**
0.324	2,169	661.1	55	16.8	2,114	644.3	47	168	74	1005	0	1294	247	1541	**771**
0.324	2,169	661.1	60	18.3	2,109	642.8	47	168	74	1191	0	1480	247	1727	**864**
0.324	2,169	661.1	65	19.8	2,104	641.3	47	168	74	1191	186	1666	247	1913	**957**
0.406	2,169	661.1	40	12.2	2,129	648.9	58	210	93	560	0	921	388	1310	**655**
0.406	2,169	661.1	45	13.7	2,124	647.4	58	210	93	793	0	1155	388	1543	**772**
0.406	2,169	661.1	50	15.2	2,119	645.9	58	210	93	1026	0	1388	388	1776	**888**
0.406	2,169	661.1	55	16.8	2,114	644.3	58	210	93	1260	0	1621	388	2010	**1005**
0.406	2,169	661.1	60	18.3	2,109	642.8	58	210	93	1493	0	1854	388	2243	**1121**
0.406	2,169	661.1	65	19.8	2,104	641.3	58	210	93	1493	233	2088	388	2476	**1238**
0.610	2,169	661.1	40	12.2	2,129	648.9	88	315	140	841	0	1384	877	2261	**1131**
0.610	2,169	661.1	45	13.7	2,124	647.4	88	315	140	1192	0	1735	877	2612	**1306**
0.610	2,169	661.1	50	15.2	2,119	645.9	88	315	140	1542	0	2085	877	2962	**1481**
0.610	2,169	661.1	55	16.8	2,114	644.3	88	315	140	1893	0	2436	877	3312	**1656**
0.610	2,169	661.1	60	18.3	2,109	642.8	88	315	140	2243	0	2786	877	3663	**1831**
0.610	2,169	661.1	65	19.8	2,104	641.3	88	315	140	2243	350	3137	877	4013	**2007**
0.762	2,169	661.1	40	12.2	2,129	648.9	109	394	175	1051	0	1729	1368	3097	**1549**
0.762	2,169	661.1	45	13.7	2,124	647.4	109	394	175	1489	0	2167	1368	3535	**1768**
0.762	2,169	661.1	50	15.2	2,119	645.9	109	394	175	1926	0	2605	1368	3973	**1986**
0.762	2,169	661.1	55	16.8	2,114	644.3	109	394	175	2364	0	3043	1368	4411	**2205**
0.762	2,169	661.1	60	18.3	2,109	642.8	109	394	175	2802	0	3480	1368	4849	**2424**
0.762	2,169	661.1	65	19.8	2,104	641.3	109	394	175	2802	438	3918	1368	5286	**2643**
0.762	2,169	661.1	70	21.3	2,099	639.8	109	394	175	2802	876	4356	1368	5724	**2862**
0.762	2,169	661.1	72	21.9	2,097	639.2	109	394	175	2802	1051	4531	1368	5899	**2950**
0.914	2,169	661.1	40	12.2	2,129	648.9	131	473	210	1260	0	2074	1968	4043	**2021**
0.914	2,169	661.1	45	13.7	2,124	647.4	131	473	210	1785	0	2599	1968	4568	**2284**
0.914	2,169	661.1	50	15.2	2,119	645.9	131	473	210	2311	0	3124	1968	5093	**2546**
0.914	2,169	661.1	55	16.8	2,114	644.3	131	473	210	2836	0	3650	1968	5618	**2809**
0.914	2,169	661.1	60	18.3	2,109	642.8	131	473	210	3361	0	4175	1968	6143	**3072**
0.914	2,169	661.1	65	19.8	2,104	641.3	131	473	210	3361	525	4700	1968	6668	**3334**
0.914	2,169	661.1	70	21.3	2,099	639.8	131	473	210	3361	1050	5225	1968	7193	**3597**
0.914	2,169	661.1	72	21.9	2,097	639.2	131	473	210	3361	1260	5435	1968	7403	**3702**

DEEP FOUNDATION PHILOSOPHY

Southern Area (see Tetra Tech's Geotechnical Report Fig 1):

The recommended soil profiles and design parameters are as follows (refer to Tetra Tech Geotechnical Report section 9.1.3):

Table 8B: Driven Steel Pile Shaft and End-Bearing Pile Parameters For Southern Area (Indicated on Figure 1)

Elevation [ft(m)]	Soil Type	Shaft Resistance			Base Resistance	
		Ultimate (kPa)	Factored for Compression (kPa)	Factored for Tension (kPa)	Ultimate (kPa)	Factored for Compression (kPa)
Above 2154 (Above 656.5)	Clay	30	15	10	-	-
2154 to 2142 (656.5 to 653.0)	Clay Till	45	22	15	-	-
2142 to 2113 (653.0 to 644.0)	Clay Till	60	30	21	800	400
Below 2113 (644.0)	Weathered Bedrock	120	60	42	3,000*	1,500

*Base resistance reduced considering bottom of pile may be partially plugged.

The corresponding calculated pile ultimate and factored compression resistances are as tabulated on the next page for the facilities located within the footprint of the Early Civil Works bathtub excavation down to EL. 2,164 ft. This includes the entire RDU process plant – please refer to the AFC civil drawings S1RD-RD-DTL-210-900-01 and -02. For pile locations where the Early Works excavation is deeper than EL. 2,164 ft (i.e. part of the Storage Tank area), the depth over which the pile shaft friction is to be neglected needs to be increased accordingly.

For the native calculation file, refer to EP Steel Pipe Pile Axial Capacities.xlsx located under: 215_STRUCTURAL\215E_ENGINEERING_DESIGN_DATA\215E_001_Job_Data_Book\Section 2 - Structural Engineering Instructions\JDB 2.04.1 Deep Foundation Philosophy\EP Pile Axial Capacities

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							Ultimate Shaft Resistance - Compression (kN)						Ultimate	Unfactored	**Factored**
Dia.	Finished Grade (El.)		Depth		Tip (El.)		Clay	Clay Till	Clay Till	Sand / Gravel	Bedrock	Total	End-bearing	Resistance	**Resistance**
(m)	(ft)	(m)	(ft)	(m)	(ft)	(m)	(kN)	(kN)	(kN)	(kN)	(kN)	(kN)	(kN)	(kN)	**(kN)**
0.273	2,169	661.1	40	12.2	2,129	648.9	63	141	204	0	0	408	47	455	**227**
0.273	2,169	661.1	45	13.7	2,124	647.4	63	141	282	0	0	486	47	533	**267**
0.273	2,169	661.1	50	15.2	2,119	645.9	63	141	361	0	0	565	47	611	**306**
0.273	2,169	661.1	55	16.8	2,114	644.3	63	141	439	0	0	643	47	690	**345**
0.273	2,169	661.1	60	18.3	2,109	642.8	63	141	455	0	125	784	176	960	**480**
0.273	2,169	661.1	65	19.8	2,104	641.3	63	141	455	0	282	941	176	1117	**558**
0.324	2,169	661.1	40	12.2	2,129	648.9	74	168	242	0	0	484	66	550	**275**
0.324	2,169	661.1	45	13.7	2,124	647.4	74	168	335	0	0	577	66	643	**322**
0.324	2,169	661.1	50	15.2	2,119	645.9	74	168	428	0	0	670	66	736	**368**
0.324	2,169	661.1	55	16.8	2,114	644.3	74	168	521	0	0	763	66	829	**415**
0.324	2,169	661.1	60	18.3	2,109	642.8	74	168	540	0	149	931	247	1178	**589**
0.324	2,169	661.1	65	19.8	2,104	641.3	74	168	540	0	335	1117	247	1364	**682**
0.406	2,169	661.1	40	12.2	2,129	648.9	93	210	303	0	0	606	104	710	**355**
0.406	2,169	661.1	45	13.7	2,124	647.4	93	210	420	0	0	723	104	827	**413**
0.406	2,169	661.1	50	15.2	2,119	645.9	93	210	537	0	0	840	104	943	**472**
0.406	2,169	661.1	55	16.8	2,114	644.3	93	210	653	0	0	956	104	1060	**530**
0.406	2,169	661.1	60	18.3	2,109	642.8	93	210	676	0	187	1166	388	1555	**777**
0.406	2,169	661.1	65	19.8	2,104	641.3	93	210	676	0	420	1400	388	1788	**894**
0.610	2,169	661.1	40	12.2	2,129	648.9	140	315	456	0	0	911	234	1145	**573**
0.610	2,169	661.1	45	13.7	2,124	647.4	140	315	631	0	0	1086	234	1320	**660**
0.610	2,169	661.1	50	15.2	2,119	645.9	140	315	806	0	0	1262	234	1495	**748**
0.610	2,169	661.1	55	16.8	2,114	644.3	140	315	981	0	0	1437	234	1671	**835**
0.610	2,169	661.1	60	18.3	2,109	642.8	140	315	1016	0	280	1752	877	2629	**1315**
0.610	2,169	661.1	65	19.8	2,104	641.3	140	315	1016	0	631	2103	877	2980	**1490**
0.762	2,169	661.1	40	12.2	2,129	648.9	175	394	569	0	0	1138	365	1503	**752**
0.762	2,169	661.1	45	13.7	2,124	647.4	175	394	788	0	0	1357	365	1722	**861**
0.762	2,169	661.1	50	15.2	2,119	645.9	175	394	1007	0	0	1576	365	1941	**970**
0.762	2,169	661.1	55	16.8	2,114	644.3	175	394	1226	0	0	1795	365	2160	**1080**
0.762	2,169	661.1	60	18.3	2,109	642.8	175	394	1270	0	350	2189	1368	3557	**1779**
0.762	2,169	661.1	65	19.8	2,104	641.3	175	394	1270	0	788	2627	1368	3995	**1997**
0.762	2,169	661.1	70	21.3	2,099	639.8	175	394	1270	0	1226	3065	1368	4433	**2216**
0.762	2,169	661.1	72	21.9	2,097	639.2	175	394	1270	0	1401	3240	1368	4608	**2304**
0.914	2,169	661.1	40	12.2	2,129	648.9	210	473	683	0	0	1365	525	1890	**945**
0.914	2,169	661.1	45	13.7	2,124	647.4	210	473	945	0	0	1628	525	2153	**1076**
0.914	2,169	661.1	50	15.2	2,119	645.9	210	473	1208	0	0	1890	525	2415	**1208**
0.914	2,169	661.1	55	16.8	2,114	644.3	210	473	1470	0	0	2153	525	2678	**1339**
0.914	2,169	661.1	60	18.3	2,109	642.8	210	473	1523	0	420	2626	1968	4594	**2297**
0.914	2,169	661.1	65	19.8	2,104	641.3	210	473	1523	0	945	3151	1968	5119	**2560**
0.914	2,169	661.1	70	21.3	2,099	639.8	210	473	1523	0	1470	3676	1968	5644	**2822**
0.914	2,169	661.1	72	21.9	2,097	639.2	210	473	1523	0	1680	3886	1968	5854	**2927**

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4.3. Single Pile Settlement

In their Geotechnical Report section 9.1.3.1, Tetra Tech state:

The estimated pile settlement for piles with 60% of the factored geotechnical resistance of the pile would be in the range of 0.5% to 1% of the pile diameter plus the elastic shortening of the pile due to applied compressive load.

For 100% of the factored geotechnical resistance of the pile, the estimated settlement would be in the range of 1% to 1.5% of the pile diameter plus the elastic shortening of the pile due to applied compressive load.

For detailed design purposes, Tetra Tech's recommendations will be implemented as follows:

- For an SLS compression load equal to 60% of the factored resistance (i.e. 30% of the ultimate pile compression resistance, Qu), the single pile settlement (excluding elastic shortening) will be assumed to be equal to 0.75% of the pile diameter. The elastic shortening can be assumed to be equal to 2.25mm.

- For an SLS compression load equal to 100% of the factored resistance (i.e. 50% of the ultimate pile compression resistance, Qu), the single pile settlement (excluding elastic shortening) will be assumed to be equal to 1.25% of the pile diameter. The elastic shortening can be assumed to be equal to 3.75mm.

- Alternatively, the elastic shortening can be estimated as follows assuming that the compression working load will be almost entirely resisted by the pile shaft friction:

  0.5 x P/A x L / E

  where P = SLS compression load (kN)
  A= Pile cross-sectional are (m^2)
  L = Pile length (m)
  E = Steel Young's Modulus (200 GPa)

- Except for the seismic load cases, the maximum SLS pile compression load should not exceed 80% (load factor = 1.25) of the maximum ULS load (also the minimum required factored geotechnical resistance). Since the geotechnical resistance factor in compression is 0.5, the maximum SLS pile compression load should not exceed 40% of the ultimate pile compression resistance (Qu). The estimated pile settlement at SLS can then therefore be interpolated between the predicted values at 30% and 50% of Qu.

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The following examples illustrate the previous recommendations:

Northern Area:

						0.3 Qu		0.5 Qu		**SLS = 0.4 Qu (say)**	
Diameter		Wall	Depth		Qu	Load	Settl.*	Load	Settl.*	Load	Settl.*
(in)	(mm)	(mm)	(ft)	(m)	(kN)	(kN)	(mm)	(kN)	(mm)	(kN)	(mm)
10 – ¾	273	9.5	50	15.2	1109	333	4.3	554	7.2	444	5.7
12 – ¾	324	9.5	50	15.2	1355	406	4.7	677	7.8	542	6.2
16	406	9.5	55	16.8	2010	603	5.3	1005	8.8	804	7.1
24	610	12.7	60	18.3	3663	1099	6.8	1831	11.4	1465	9.1
24	610	12.7	65	19.8	4013	1204	6.8	2007	11.4	1605	9.1
30	762	15.9	65	19.8	5286	1586	8.0	2643	13.3	2115	10.6
30	762	15.9	70	21.3	5724	1717	8.0	2862	13.3	2290	10.6
36	914	19.0	65	19.8	6668	2000	9.1	3334	15.2	2667	12.1
36	914	19.0	70	21.3	7193	2158	9.1	3597	15.2	2877	12.1

(*) Including elastic shortening assumed to be 2.25mm at 0.3 Qu and 3.75mm at 0.5 Qu.

Southern Area:

						0.3 Qu		0.5 Qu		**SLS = 0.4 Qu (say)**	
Diameter		Wall	Depth		Qu	Load	Settl.*	Load	Settl.*	Load	Settl.*
(in)	(mm)	(mm)	(ft)	(m)	(kN)	(kN)	(mm)	(kN)	(mm)	(kN)	(mm)
10 – ¾	273	9.5	50	15.2	611	183	4.3	306	7.2	245	5.7
12 – ¾	324	9.5	50	15.2	736	221	4.7	368	7.8	294	6.2
16	406	9.5	55	16.8	1060	318	5.3	530	8.8	424	7.1
24	610	12.7	60	18.3	2629	789	6.8	1315	11.4	1052	9.1
24	610	12.7	65	19.8	2980	894	6.8	1490	11.4	1192	9.1
30	762	15.9	65	19.8	3995	1198	8.0	1997	13.3	1598	10.6
30	762	15.9	70	21.3	4433	1330	8.0	2216	13.3	1773	10.6
36	914	19.0	65	19.8	5119	1536	9.1	2560	15.2	2048	12.1
36	914	19.0	70	21.3	5644	1693	9.1	2822	15.2	2258	12.1

(*) Including elastic shortening assumed to be 2.25mm at 0.3 Qu and 3.75mm at 0.5 Qu.

Unless agreed otherwise with the CSA Lead Engineer, piled foundations for structures and equipment shall be designed not to exceed a total settlement of ½” (12.7mm) and a differential settlement between adjacent foundations of ¼” (6.4mm) under working loads. Pile group effects are required to be taken into consideration in the design.

4.4. Negative Skin Friction

Per Tetra Tech recommendations in section 9.1.3.2 of the Geotechnical Report, negative skin friction (downdrag) is only to be considered for piles near the storage tanks where the depth of select engineering fill is expected to be more than 2m (6.6 ft). A negative skin friction of 40 kPa shall be assumed over the full thickness of the select engineering fill placed during the Civil Early Works to calculate the total downdrag force.

Piles shall be checked structurally when subjected to concurrent permanent (i.e. dead) loads and downdrag. A load factor of 1.25 shall be used to derive the factored downdrag force.

Pile settlement also needs to be checked for the downdrag condition. Tetra Tech will be requested to perform additional analyses for the piles located in close proximity to the storage tanks.

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4.5. Pile Spacing

Steel pipe piles shall be installed at a minimum pile spacing of 3 diameters centre-to-centre. A reduced pile spacing of 2.5 diameters centre-to-centre should only be used in exceptional circumstances where the layout of the facilities and loading conditions provide additional design constraints and acknowledging the greater potential for pile interaction and heave during driving.

4.6. Group Efficiency Factors for Ultimate Axial Resistance

Tetra Tech's Geotechnical Report is silent on this topic. Since the piles will be founded in either dense sands/gravels, very stiff to hard glacial till or weathered bedrock, the factored geotechnical resistance calculated from the design parameters for individual piles does not need to be reduced for pile groups. As such, the ultimate compression resistance of a pile group can be assumed to be the sum of the ultimate compression capacities of the individual piles in the considered group.

4.7. Pile Group Settlement

Settlement of pile groups will be larger than the settlement of individual piles under the same average pile loading. For the estimation of pile group settlements, group multiplication factors (Rs) are to be applied to the estimated single pile settlement under the same average load.

In the Geotechnical Report section 9.1.4.1, Tetra Tech provided the following recommendations:

The recommendations from non-linear theories are presented as follows. For design consideration, the effect of group action can be expressed as settlement ratio, R_s, where:

$$R_s = \frac{\text{Average group settlement considering load per pile}}{\text{Settlement of a single pile under the same average load}}$$

Group effects on settlement are considered to diminish at a pile spacing of at least seven pile diameters (i.e., R_s = 1.0 at seven pile diameters). For typical centre to centre pile spacing within a group of approximately 3, 4, 6 and 8 pile diameters, with pile tops connected within a rigid frame or pile cap, the following settlement ratios are recommended. The following ratios may be used for piles up to 30 m in length.

Table 10: Settlement Ratios

Group Size	Settlement Ratio, Rs				
	2.5D	**3D**	**4D**	**6D**	**7D**
1 x 2	1.4	1.3	1.22	1.07	1.0
2 x 2	2.0	1.6	1.45	1.15	1.0
3 x 3	2.5	2.0	1.75	1.25	1.0
4 x 4	3.0	2.4	2.05	1.35	1.0
5 x 5	3.5	2.8	2.35	1.45	1.0
6 x 6	4.0	3.2	2.65	1.55	1.0
10 x 10	6.0	4.8	3.85	1.95	1.0

The estimation of the settlement of a pile group under static loads is undertaken as follows.

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Estimated pile group settlement = Estimated settlement of a single pile due to the average imposed SLS permanent (i.e. dead) compression load times the applicable vertical group multiplication factor (Rs) for the group size and configuration.

The value of Rs can be interpolated between the values provided in the above Table 10.

4.8. Single Pile Ultimate Tension Resistance (excl. frost uplift)

Based on the recommendations from Tetra Tech, a single pile ultimate resistance against tension (excl. frost uplift) can be estimated from the following equation:

Qsu = qsu x As

where: Q_{su} = Ultimate geotechnical pile tension resistance (kN)
qsu = ultimate skin friction (kPa)
As = shaft area (m^2)

For the recommended design soil profiles and ultimate skin frictions, refer to section 4.2 herein. **Note the recommendation to neglect the skin friction within the select engineering fill placed during the Early Civil Works or within all soils above EL 2162ft, whichever is the most onerous, for the entire site.**

A geotechnical resistance factor (Φ) of 0.35 (excl. frost) in tension is recommended based on the performed pile PDA tests (refer to Tetra Tech's Geotechnical Report section 9.1.3).

The calculated factored tension resistances for 273mm to 914mm dia. piles are tabulated overleaf for the facilities located within the footprint of the Early Civil Works bathtub excavation down to EL. 2,164 ft. For the bathtub excavation limits, please refer to the AFC civil drawings S1RD-RD-DTL-210-900-01 and -02.

For the native calculation file, refer to EP Steel Pipe Pile Axial Capacities.xlsx located under: 215_STRUCTURAL\215E_ENGINEERING_DESIGN_DATA\215E_001_Job_Data_Book\Section 2 - Structural Engineering Instructions\JDB 2.04.1 Deep Foundation Philosophy\EP Pile Axial Capacities

For pile locations where the Early Works excavation was deeper than EL. 2,164 ft (i.e. part of the Storage Tank area), the depth over which the pile shaft friction is to be neglected needs to be increased accordingly.

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Northern Area (see Tetra Tech's Geotechnical Report Fig 1):

							Ultimate Shaft Resistance - Tension (kN)						**Factored**
Dia.	Finished Grade (El.)		Depth		Tip (El.)		Clay	Clay Till	Clay Till	Sand / Gravel	Bedrock	Total	**Resistance**
(m)	(ft)	(m)	(ft)	(m)	(ft)	(m)	(kN)	(kN)	(kN)	(kN)	(kN)	(kN)	**(kN)**
0.273	2,169	661.1	40	12.2	2,129	648.9	39	141	63	376	0	620	**217**
0.273	2,169	661.1	45	13.7	2,124	647.4	39	141	63	533	0	776	**272**
0.273	2,169	661.1	50	15.2	2,119	645.9	39	141	63	690	0	933	**327**
0.273	2,169	661.1	55	16.8	2,114	644.3	39	141	63	847	0	1090	**382**
0.273	2,169	661.1	60	18.3	2,109	642.8	39	141	63	1004	0	1247	**436**
0.273	2,169	661.1	65	19.8	2,104	641.3	39	141	63	1004	157	1404	**491**
0.324	2,169	661.1	40	12.2	2,129	648.9	47	168	74	447	0	735	**257**
0.324	2,169	661.1	45	13.7	2,124	647.4	47	168	74	633	0	921	**323**
0.324	2,169	661.1	50	15.2	2,119	645.9	47	168	74	819	0	1108	**388**
0.324	2,169	661.1	55	16.8	2,114	644.3	47	168	74	1005	0	1294	**453**
0.324	2,169	661.1	60	18.3	2,109	642.8	47	168	74	1191	0	1480	**518**
0.324	2,169	661.1	65	19.8	2,104	641.3	47	168	74	1191	186	1666	**583**
0.406	2,169	661.1	40	12.2	2,129	648.9	58	210	93	560	0	921	**322**
0.406	2,169	661.1	45	13.7	2,124	647.4	58	210	93	793	0	1155	**404**
0.406	2,169	661.1	50	15.2	2,119	645.9	58	210	93	1026	0	1388	**486**
0.406	2,169	661.1	55	16.8	2,114	644.3	58	210	93	1260	0	1621	**567**
0.406	2,169	661.1	60	18.3	2,109	642.8	58	210	93	1493	0	1854	**649**
0.406	2,169	661.1	65	19.8	2,104	641.3	58	210	93	1493	233	2088	**731**
0.610	2,169	661.1	40	12.2	2,129	648.9	88	315	140	841	0	1384	**485**
0.610	2,169	661.1	45	13.7	2,124	647.4	88	315	140	1192	0	1735	**607**
0.610	2,169	661.1	50	15.2	2,119	645.9	88	315	140	1542	0	2085	**730**
0.610	2,169	661.1	55	16.8	2,114	644.3	88	315	140	1893	0	2436	**853**
0.610	2,169	661.1	60	18.3	2,109	642.8	88	315	140	2243	0	2786	**975**
0.610	2,169	661.1	65	19.8	2,104	641.3	88	315	140	2243	350	3137	**1098**
0.762	2,169	661.1	40	12.2	2,129	648.9	109	394	175	1051	0	1729	**605**
0.762	2,169	661.1	45	13.7	2,124	647.4	109	394	175	1489	0	2167	**758**
0.762	2,169	661.1	50	15.2	2,119	645.9	109	394	175	1926	0	2605	**912**
0.762	2,169	661.1	55	16.8	2,114	644.3	109	394	175	2364	0	3043	**1065**
0.762	2,169	661.1	60	18.3	2,109	642.8	109	394	175	2802	0	3480	**1218**
0.762	2,169	661.1	65	19.8	2,104	641.3	109	394	175	2802	438	3918	**1371**
0.762	2,169	661.1	70	21.3	2,099	639.8	109	394	175	2802	876	4356	**1525**
0.762	2,169	661.1	72	21.9	2,097	639.2	109	394	175	2802	1051	4531	**1586**
0.914	2,169	661.1	40	12.2	2,129	648.9	131	473	210	1260	0	2074	**726**
0.914	2,169	661.1	45	13.7	2,124	647.4	131	473	210	1785	0	2599	**910**
0.914	2,169	661.1	50	15.2	2,119	645.9	131	473	210	2311	0	3124	**1094**
0.914	2,169	661.1	55	16.8	2,114	644.3	131	473	210	2836	0	3650	**1277**
0.914	2,169	661.1	60	18.3	2,109	642.8	131	473	210	3361	0	4175	**1461**
0.914	2,169	661.1	65	19.8	2,104	641.3	131	473	210	3361	525	4700	**1645**
0.914	2,169	661.1	70	21.3	2,099	639.8	131	473	210	3361	1050	5225	**1829**
0.914	2,169	661.1	72	21.9	2,097	639.2	131	473	210	3361	1260	5435	**1902**

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Southern Area (see Tetra Tech's Geotechnical Report Fig 1):

							Ultimate Shaft Resistance - Tension (kN)						**Factored**
Dia.	Finished Grade (El.)		Depth		Tip (El.)		Clay	Clay Till	Clay Till	Sand / Gravel	Bedrock	Total	**Resistance**
(m)	(ft)	(m)	(ft)	(m)	(ft)	(m)	(kN)	(kN)	(kN)	(kN)	(kN)	(kN)	**(kN)**
0.273	2,169	661.1	40	12.2	2,129	648.9	63	141	204	0	0	408	**143**
0.273	2,169	661.1	45	13.7	2,124	647.4	63	141	282	0	0	486	**170**
0.273	2,169	661.1	50	15.2	2,119	645.9	63	141	361	0	0	565	**198**
0.273	2,169	661.1	55	16.8	2,114	644.3	63	141	439	0	0	643	**225**
0.273	2,169	661.1	60	18.3	2,109	642.8	63	141	455	0	125	784	**274**
0.273	2,169	661.1	65	19.8	2,104	641.3	63	141	455	0	282	941	**329**
0.324	2,169	661.1	40	12.2	2,129	648.9	74	168	242	0	0	484	**169**
0.324	2,169	661.1	45	13.7	2,124	647.4	74	168	335	0	0	577	**202**
0.324	2,169	661.1	50	15.2	2,119	645.9	74	168	428	0	0	670	**235**
0.324	2,169	661.1	55	16.8	2,114	644.3	74	168	521	0	0	763	**267**
0.324	2,169	661.1	60	18.3	2,109	642.8	74	168	540	0	149	931	**326**
0.324	2,169	661.1	65	19.8	2,104	641.3	74	168	540	0	335	1117	**391**
0.406	2,169	661.1	40	12.2	2,129	648.9	93	210	303	0	0	606	**212**
0.406	2,169	661.1	45	13.7	2,124	647.4	93	210	420	0	0	723	**253**
0.406	2,169	661.1	50	15.2	2,119	645.9	93	210	537	0	0	840	**294**
0.406	2,169	661.1	55	16.8	2,114	644.3	93	210	653	0	0	956	**335**
0.406	2,169	661.1	60	18.3	2,109	642.8	93	210	676	0	187	1166	**408**
0.406	2,169	661.1	65	19.8	2,104	641.3	93	210	676	0	420	1400	**490**
0.610	2,169	661.1	40	12.2	2,129	648.9	140	315	456	0	0	911	**319**
0.610	2,169	661.1	45	13.7	2,124	647.4	140	315	631	0	0	1086	**380**
0.610	2,169	661.1	50	15.2	2,119	645.9	140	315	806	0	0	1262	**442**
0.610	2,169	661.1	55	16.8	2,114	644.3	140	315	981	0	0	1437	**503**
0.610	2,169	661.1	60	18.3	2,109	642.8	140	315	1016	0	280	1752	**613**
0.610	2,169	661.1	65	19.8	2,104	641.3	140	315	1016	0	631	2103	**736**
0.762	2,169	661.1	40	12.2	2,129	648.9	175	394	569	0	0	1138	**398**
0.762	2,169	661.1	45	13.7	2,124	647.4	175	394	788	0	0	1357	**475**
0.762	2,169	661.1	50	15.2	2,119	645.9	175	394	1007	0	0	1576	**552**
0.762	2,169	661.1	55	16.8	2,114	644.3	175	394	1226	0	0	1795	**628**
0.762	2,169	661.1	60	18.3	2,109	642.8	175	394	1270	0	350	2189	**766**
0.762	2,169	661.1	65	19.8	2,104	641.3	175	394	1270	0	788	2627	**919**
0.762	2,169	661.1	70	21.3	2,099	639.8	175	394	1270	0	1226	3065	**1073**
0.762	2,169	661.1	72	21.9	2,097	639.2	175	394	1270	0	1401	3240	**1134**
0.914	2,169	661.1	40	12.2	2,129	648.9	210	473	683	0	0	1365	**478**
0.914	2,169	661.1	45	13.7	2,124	647.4	210	473	945	0	0	1628	**570**
0.914	2,169	661.1	50	15.2	2,119	645.9	210	473	1208	0	0	1890	**662**
0.914	2,169	661.1	55	16.8	2,114	644.3	210	473	1470	0	0	2153	**754**
0.914	2,169	661.1	60	18.3	2,109	642.8	210	473	1523	0	420	2626	**919**
0.914	2,169	661.1	65	19.8	2,104	641.3	210	473	1523	0	945	3151	**1103**
0.914	2,169	661.1	70	21.3	2,099	639.8	210	473	1523	0	1470	3676	**1287**
0.914	2,169	661.1	72	21.9	2,097	639.2	210	473	1523	0	1680	3886	**1360**

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4.9. Single Pile Lateral Resistance (SLS)

The allowable pile lateral deflection limit (SLS) at grade under static loads shall be limited to **12mm** for all pile sizes unless approved otherwise by the CSA Lead Engineer. This geotechnical limitation is not required to be checked under seismic loading conditions. The permissible pile lateral deflection may also be tighter if dictated otherwise by the sensitivity of the structure / equipment being supported.

Tetra Tech provided recommended design soil parameters in their Geotechnical Report section 9.1.3.3 Table 9a:

Table 9A: Lateral Pile Capacity Analysis Parameters for Unfrozen Soils

Soil Unit	Model	Unit Weight (kN/m³)	Undrained Shear Strength (kPa)	Angle of Internal Friction (°)	Strain Factor (ε_{50} or k_{rm})
Granular Structural Fill (min 98% SPMDD)	Sand (Reese)	21	-	36	33,900 kPa/m
Select Fill (min 98% SPMDD)	Stiff Clay (Reese)	20	75	-	0.007
Native Clay	Stiff Clay (Reese)	17	50	-	0.008
Clay Till	Stiff Clay (Reese)	20	75	-	0.007
Sand or Sand and Gravel	Sand (Reese)	20	-	36	33,900 kPa/m
Weathered Bedrock	Stiff Clay (Reese)	21	200	-	0.005

For detailed design purposes and based on the Tetra Tech recommended soil profiles for the Northern & Southern areas (refer to section 4.2 herein), the following design soil profile was assumed herein.

Geological Unit	Elevation m (ft)		Thickness (m)
	Top	Bottom	
Granular Select fill* (min 98% SPMDD)	661 (2169)	659.5 (2164)	1.5
Native Clay	659.5 (2164)	656.5 (2154)	3.0
Clay Till	656.5 (2154)	643 (2109)	13.5
Weathered Bedrock	2109 (643)		

(*) Worst case for select fill (i.e. granular vs cohesive)

The groundwater table was conservatively assumed to be at 1.5m below FGL (i.e. EL 659.5m or 2164 ft) in accordance with Tetra Tech's recommendations in the Geotechnical Report section 4.2.

Tetra Tech also recommended LPile soil parameters for frozen soil conditions (refer to Table 9B in Section 9.1.3.3 of their geotechnical report). Specific LPile runs with frozen soils can be performed as required.

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Using Tetra Tech's recommended soil parameters, a series of single pile lateral analyses were carried out using LPile v2019.11.08 for both free and fixed head conditions based on a Finished Grade Elevation of EL. 661m or 2169 ft.
The following free-head scenarios were assumed (see **Appendix B**):

- Pile head at 500mm above finished grade assumed at EL 2169FT (or EL 661m);
- 273mm, 324mm, 406mm, 610mm, 762mm & 914mm dia. piles;
- Corrosion allowance on pile wall thickness (i.e. 1.5mm);
- Applied loads are static (i.e. not cyclic);
- Vertical compression loads have a 75mm eccentricity (to account for the permitted "as-built" tolerance on the pile location).

The following free-head scenarios were assumed (see **Appendix C**):

- Pinned pile head at underside of concrete pilecap at EL 2164 FT (or EL 659.5m);
- 406mm, 610mm, 762mm & 914mm dia. piles;
- Corrosion allowance on pile wall thickness (i.e. 1.5mm);
- Applied loads are static (i.e. not cyclic).
- No eccentricity as concrete pilecap eliminates the effective eccentricity on the piles.

The following fixed-head scenarios were assumed (see **Appendix D**):

- Fixed pile head at underside of concrete pilecap at EL 2164 FT (or EL 659.5m);
- 406mm, 610mm, 762mm & 914mm dia. piles;
- Corrosion allowance on pile wall thickness (i.e. 1.5mm);
- Applied loads are static (i.e. not cyclic).
- No eccentricity as concrete pilecap eliminates the effective eccentricity on the piles.

Note that achieving full fixity and resisting the maximum developing bending moment at the head of the piles can be difficult. As such, it is preferred to assume free head conditions for pile groups with a reinforced concrete pilecap and utilize Nelson studs (shear connectors) per standard drawing S1RD-RD-STD-215-001-01 pile cap detail Type B.

All referenced LPile files are located under:
215_STRUCTURAL\215E_ENGINEERING_DESIGN_DATA\215E_001_Job_Data_Book\Section 2 - Structural Engineering Instructions\JDB 2.04.1 Deep Foundation Philosophy\LPile runs
These LPile files can be used by the Structural Engineer to perform his own pile lateral analyses for his specific piling configurations.

Pile group effects are also required to be taken into consideration as outlined in the next section.

4.10. Pile Group Lateral Resistance (SLS)

The estimation of the deflection of a pile group under static loads is undertaken as shown overleaf.

<u>Method # 1:</u> The following p-y curve multipliers for group effects were recommended by Tetra Tech in the Geotechnical Report section 9.1.4.2 and can be used directly in LPile to estimate the behaviour of pile groups. The structural engineer will need to include the p-y multiplier in his LPile analysis of a single pile (subject to the average SLS and/or ULS lateral loading) in order to reduce the lateral soil resistance and obtain predicted deflections and max. bending moments that would include group effects.

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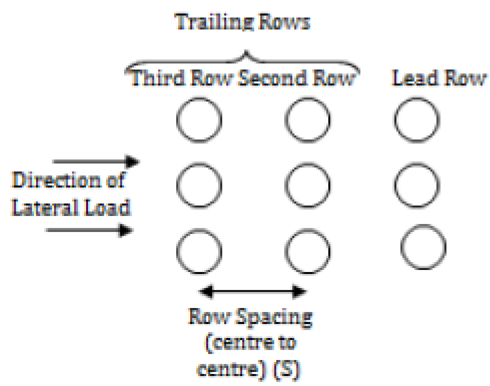

Pile Group Nomenclature for Lateral Loads

Table 11: p-y Curve Multipliers for Group Effects

Pile Row	Row Spacing					
	2.5D	3D	4D	5D	6D	8D
Lead	0.7	0.8	0.85	0.9	1	1
Second	0.5	0.6	0.7	0.8	0.9	1
Third and Subsequent Rows	0.3	0.4	0.6	0.7	0.8	1

Method # 2: Group multiplication factors are commonly used to estimate the deflection of a pile group from the behaviour of a single pile subjected to the same average lateral load per pile as in the group. These group multiplication factors can be conservatively estimated from the recommended p-y multipliers.

Estimated pile group deflection = Estimated deflection of a single pile (under the average imposed service lateral load) times the applicable lateral group multiplication factor for the given group size and configuration.

Based on the recommendations from Tetra Tech for the p-y multipliers, the following lateral group multiplication factors were calculated by Fluor (as the inverse of the average p-y multiplier for the group) and can be used for design purposes:

Piles in Group	Lateral Group Multiplication Factor					
	2.5D	3D	4D	5D	6D	8D
2x1 (side by side)	1.43	1.25	1.18	1.11	1.00	1.00
2x1	1.67	1.43	1.29	1.18	1.05	1.00
2x2	n/a	1.43	1.29	1.18	1.05	1.00
3x3	n/a	1.67	1.40	1.25	1.11	1.00
4X4	n/a	1.82	1.45	1.29	1.15	1.00
5X5	n/a	1.92	1.49	1.32	1.16	1.00
6X6	n/a	2.00	1.52	1.33	1.18	1.00
8X8	n/a	2.10	1.55	1.35	1.20	1.00

DEEP FOUNDATION PHILOSOPHY

4.11. Pile Lateral Resistance (ULS)

The Structural Engineer must check that the structural capacity of the piles (including the 1.5mm corrosion allowance) is not exceeded at ULS (both static and seismic conditions). Typically this is not the case since the maximum imposed stress conditions typically occur during driving.

However, structural checks are still required and can be performed for the governing ULS load conditions using LPile (refer to section 4.9).

The maximum pile bending moment at ULS in a group configuration can be estimated by:

- Method 1: Running LPile with the recommended average p-y multiplier for the given group configuration and with the governing average ULS axial & lateral loads per pile.

- Method 2: Multiplying the LPile calculated max. bending moment for a single pile under the same governing average ULS axial & lateral loads per pile by the applicable group multiplication factor (refer to section 4.10).

5.0 HELICAL PILES

5.1. Introduction

Helical piles can be considered for light and non-settlement sensitive structures only (refer to S1RD-PDR-006).

Helical piles are typically designed, supplied and installed by a Specialist Contractor. As such, Fluor would only typically develop conceptual designs for helical piles.

Specification # S1RD-RD-SPC-215-010-01 was prepared to cover specifically helical pile requirements on the Project. Per this specification, please note that the shaft outside diameter of an helical pile shall not exceed 10 – ¾ in (273mm) unless approved otherwise in writing by IOL.

In Fluor's experience, helical piles with a diameter comprised between 8 – 5/8 in (219mm) and 16 in (406mm) are commonly used in our industry. The helix diameter should not typically exceed 2.5 times the shaft diameter as required by the Project specification.

The following summary table provides potential helical pile sizes that could be considered as a guide by the structural engineer on the Project:

Pipe OD (in)	Pipe OD (mm)	Min. Wall Thickness (mm)	Helix Dia. (in)	Helix Dia. (mm)
8 - 5/8	219	8.2	16 - 20	406 - 508
10 - 3/4	273	9.3	20 - 26	508 - 660
12 - 3/4	323	9.5	22 - 30	559 - 762
14	355	9.5	24 - 34	610 - 864
16	406	9.5	28 - 40	711 - 1016

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5.2. Compression & Tension Resistance

In their Geotechnical Report section 9.1.8, Tetra Tech highlights the potential risk of practical refusal in the shallow sands / gravels and provide soil design parameters in Tables 12A & 12B (also included below) for the Northern and Southern areas (see Fig 1 in Appendix A).

However, Tetra Tech do not provide shaft and end-bearing resistance values as well as complete details of the methodology to be followed for the conceptual design of helical piles depending on the number of helices and their configuration.

As such, the preliminary sizing of helical piles during the EP phase shall preferably be performed by, or under the direct supervision of, Chris Palanque (Geotechnical SME) using the loading conditions provided by the Structural Engineer.

Table 12A: Geotechnical Parameters for Helical Piles – ULS For Northern Area (Indicated on Figure 1)

Elevation [ft(m)]	Soil Type	Bulk Unit Weight (kN/m^3)	Angle of Internal Friction (°)	Undrained Shear Strength (kPa)
Above 2157 657.7	Native Clay	17	20	50
2157 to 2141 (657.5 to 652.6)	Clay Till	20	26	75
2141 to 2109 (652.6 to 643.0)	Sand and gravel**	21	36	-
Below 2109 (643.0)	Weathered Bedrock	21	24	200

**May be difficult to penetrate this layer (N>50).

Table 12B: Geotechnical Parameters for Helical Piles – ULS For Southern Area (Indicated on Figure 1)

Elevation [ft(m)]	Soil Type	Bulk Unit Weight (kN/m^3)	Angle of Internal Friction (°)	Undrained Shear Strength (kPa)
Above 2154 656.5	Native Clay	17	20	50
2154 to 2113 (656.5 to 644.0)	Clay Till	20	26	75
Below 2113 (644.0)	Weathered Bedrock	21	24	200

5.3. Lateral Resistance

The lateral resistance checks (both SLS & ULS) for an helical pile are performed in the same fashion as for a driven pipe pile (refer to sections 4.9 to 4.11). However, due to the potential soil disturbing effects during installation, it is not uncommon to assume reduced soil shear strength parameters.

For conceptual design purposes, the lateral analyses of helical piles shall use the soil parameters recommended by Tetra Tech in the above sections 4.9 and 4.10.

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As for the axial resistance, it is preferable that the LPile lateral analyses be performed by, or under the direct supervision of, Chris Palanque (Geotechnical SME).

6.0 FROST UPLIFT

6.1. Frost Depth

Tetra Tech recommended a frost depth of 7.9 ft (2.4m) for clay soils (i.e. native clays and clay fill) and 11 ft (3.3m) for granular backfill based on 1:50 year event (refer to the Geotechnical Report section 8.0). For the select engineered fill placed during the early civil works, the recommended frost penetration depth is 7.9 ft (2.4m):

The near surface soils are considered to have medium to high frost susceptibility. With a mean annual AFI (Annual Freezing Index) of 1750°C-days (corrected to a design freezing index of 2400 °C-days) for a 1:50 year return period, the frost penetration in clay soils is estimated to be 7.9 ft. (2.4 m). The frost penetration in granular backfill is estimated to be 11 ft. (3.3 m). For the select engineered fill placed during the early civil works, the maximum frost penetration depth is estimated to be 7.9 ft (2.4 m).

In the RDU unit and the adjacent Tank Storage Farm, an approximately 6 ft deep bathtub excavation from the originally existing grade (i.e. approx. Plant EL 2,170 ft) down to Plant EL 2,164 ft will be backfilled with select engineering fill. An additional central excavation below the bathtub down to EL 2,157'- 6" will also be carried out in the Tank Storage Farm and will be backfilled with select engineering fill.

A review of all the available site boreholes (i.e. BH01 to BH06) show clay or clay fill materials below EL 2,164 ft down to a depth of 7.9 ft (2.4m) below FGL.

The design frost depth (1:50 year) can therefore be assumed to be 7.9ft (2.4m) per Tetra Tech's recommendations.

The underside of piled concrete foundations may not be set below the design frost depth. In addition, for large concrete foundations, the depth of frost penetration may increase due to the lower thermal resistance of the concrete. It is therefore recommended to place a frost cushion material at the underside of all non-insulated pilecaps where frost heave concerns exist (see section 6.3 herein).

6.2. Adfreeze

Adfreeze structural forces shall be considered alongside all exposed vertical surfaces (e.g. pile shaft, pilecaps, grade beams) located within the depth of frost penetration.

The adfreeze uplift stress shall be assumed to be **100kPa** for detailed design purposes around **steel** pile shafts over the design frost depth (refer to Tetra Tech's report sections 8.1 & 9.1.3).

The use of a suitable bond breaker (e.g. a single layer of 30 mil PVC or LDPE) should be considered on all vertical faces of buried concrete pilecaps and grade beams to reduce adfreeze forces. With reference to Tetra Tech's report section 8.2, a reduced adfreeze value of **32.5 kPa** (i.e. 50% of 65 kPa) is recommended for a single layer of 30 mil PVC or LDPE. If a bond-breaker is not used, the adfreeze uplift stress around buried **concrete** features shall be assumed to be

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65kPa over the design frost depth as recommended by Tetra Tech in their report section 8.1, and per the CFEM 4th edition section 13.5.1.

6.3. Frost Heave

In the Geotechnical Report Section 8.0, Tetra Tech recommended a design frost heave of 50mm:

For design purposes, a frost heave value of 50 mm is recommended, based on the "select engineered fill" being used for the backfill for the early civil works.

Where frost heave concerns exist, either a layer of Styrofoam insulation or a fully rebounding frost cushion product made from Ethafoam such as Nordic 100, 200 or 400 manufactured by Norseman (or equivalent) is recommended. The use of a non-rebounding (e.g. crushable) void form is a concern due to the potentially high water table. The frost heave force to be considered in the design of the frost cushion will need to be determined based on the adopted product type (e.g. Nordic 100, 200 or 400) and thickness using a frost heave design value of 50mm. For Norseman Nordic frost cushion, a minimum thickness of 200mm is to be considered with compressive stress vs strain curves being available for use under the following project directory folder:

215_STRUCTURAL\215E_ENGINEERING_DESIGN_DATA\215E_006_R_Geotechnical_Rpt\Norseman Frost Cushion

6.4. Frost Uplift Checks

The following check shall be performed for piled foundations to ensure an adequate factor of safety against uplift due to frost effects is available.

Case 1 – Single Pile Subject to Adfreeze (i.e. No Pilecap)

$\boldsymbol{\Phi R_n \geq \alpha S_n}$ [Ref. Canadian Foundation Engineering Manual (CFEM)]

where:
R_n = Ultimate pile tension resistance against frost effects (kN)
Φ = Geotechnical resistance factor = 0.8 (see Tetra Tech's Geotechnical Report section 9.1.3)
S_n = Adfreeze on single pile shaft (kN)
α = Load factor on adfreeze = 1.0

Case 2 – Piled Foundation (i.e. with a pilecap) Subject to Adfreeze & Frost Heave

$\boldsymbol{\Phi R_n \times N \geq (\alpha_1 \times Adf_1 \times N) + (\alpha_2 \times Adf_2) + (\alpha_3 \times Fh) - (\alpha_4 \times D_1) - (\alpha_5 \times D_2)}$

where:
R_n = Ultimate pile tension resistance against frost effects (kN)
Φ = Geotechnical resistance factor = 0.8
N = number of piles
$\alpha_1 = \alpha_2 = \alpha_3$ = Load factor on frost forces = 1.0
Adf_1 = Adfreeze on single pile shaft (kN)
Adf_2 = Adfreeze on pilecap (kN)
Fh = Frost heave load on underside of pilecap (kN)
$\alpha_4 = \alpha_5$ = Load factor on beneficial sustained structural forces = 0.9
D_1 = Pilecap self-weight (kN)
D_2 = Min. sustained structural load imposed on pilecap (as applicable) (kN)

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Case 3 – Piled Foundation (i.e. with a pilecap) Subject to Adfreeze, Frost Heave & Sustained Net Tensile Structural Load

$$\mathbf{R_n \times N \geq [(\alpha_1 \times Adf_1 \times N) + (\alpha_2 \times Adf_2) + (\alpha_3 \times Fh) - (\alpha_4 \times D)] / \Phi_1 + (\alpha_6 \times T)/ \Phi_2}$$

where:
R_n = Ultimate pile tension resistance against frost effects (kN)
Φ_1 = Geotechnical resistance factor for frost effects = 0.8
Φ_2 = Geotechnical resistance factor for sustained net tensile load = 0.35
N = number of piles
Adf_1 = Adfreeze on single pile shaft (kN)
Adf_2 = Adfreeze on pilecap (kN)
Fh = Frost heave load on underside of pilecap (kN)
$\alpha_1 = \alpha_2 = \alpha_3$ = Load factor on frost forces = 1.0
D = Pilecap self-weight (kN)
T = Sustained net tensile load applied to pilecap (kN)
α_4 = Load factor on pilecap dead weight = 0.9
α_6 = Load factor on sustained net tensile structural load (to be determined by structural engineer)

Note: The above formulae are applicable to cases where the pile group can reasonably be expected to uniformly share the resistance to uplift on the pile cap. Engineers are responsible for determining the applicability of this method to the item they are designing and, as required, performing an appropriate evaluation for other cases.

6.5. Minimum Pile Embedment (Frost)

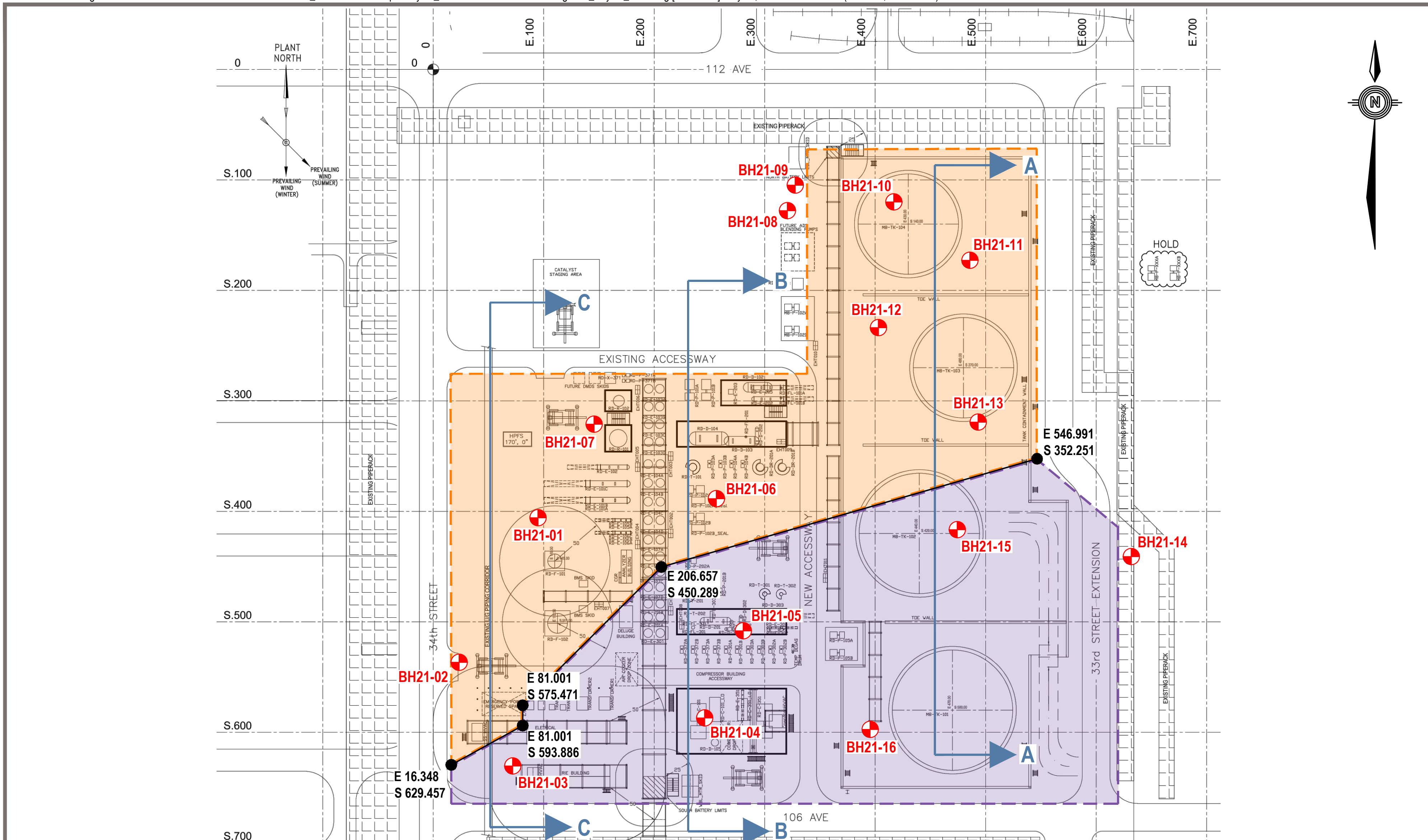
With reference to the excel file EP Steel Pipe Pile Axial Capacities.xlsx prepared by Fluor, the minimum required embedment depth of a single pile projecting above finished grade to resist frost uplift forces (see section 6.4 case 1) is 29 ft (8.9m) for the Northern Area and 31 ft (9.5m) for the Southern Area based on an assumed frost depth of 7.9 ft (2.4m) (see section 6.1) and an adfreeze stress of 100 kPa (see section 6.2).

For simplicity, it is recommended for detailed design purposes to embed all piles projecting above finished grade a **minimum of 33 ft (10m)** based on a frost depth of 7.9 ft (2.4m).

APPENDIX A

Definition of Northern & Southern Areas

Q:\Edmonton\Drafting\PROJECTS\704-ENG.EGEO\ENG.EGEO04009-02_IOL Shred Desktop Study\02_Acad\ENG.EGEO04009-02 Figure 1_July 27_2022.dwg [FIGURE 1] July 27, 2022 - 11:32:59 am (BY: DAS, DEBASHIS)



LEGEND:

- 2021 BOREHOLE LOCATION (APPROXIMATE)
- NORTHERN AREA FOR PILE DESIGN
- SOUTHERN AREA FOR PILE DESIGN
- SECTION

NOTES:

1) DRAWING NOT TO SCALE
2) DRAWING TAKEN FROM IOL DRAWING NO. 119500 REV C, DATED JANUARY 27, 2022 VENDOR DRAWING NO. 142-KE-500

CLIENT

Imperial

TETRA TECH

SHRED PROJECT
STRATHCONA REFINERY, EDMONTON, ALBERTA

SITE PLAN SHOWING BOREHOLE AND SECTION LOCATIONS

PROJECT NO.	DWN	CKD	REV
SWM.SWOP04009-02	DBD	VO	0

OFFICE	DATE
EDM	July 2022

Figure 1

DEEP FOUNDATION PHILOSOPHY

APPENDIX B

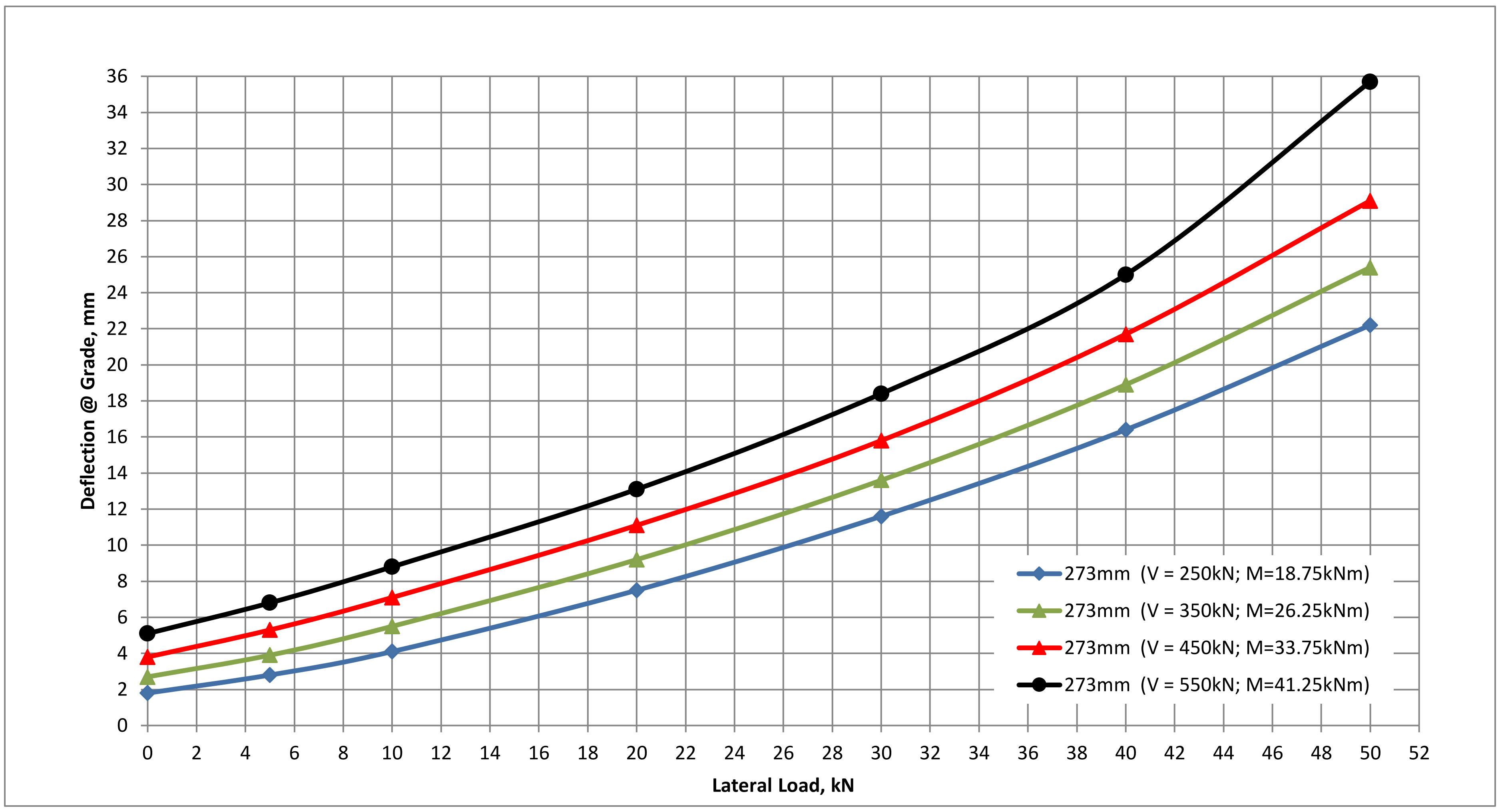
Lateral Analyses of Single Piles (free head) Pile Head at 0.5m above Finished Grade

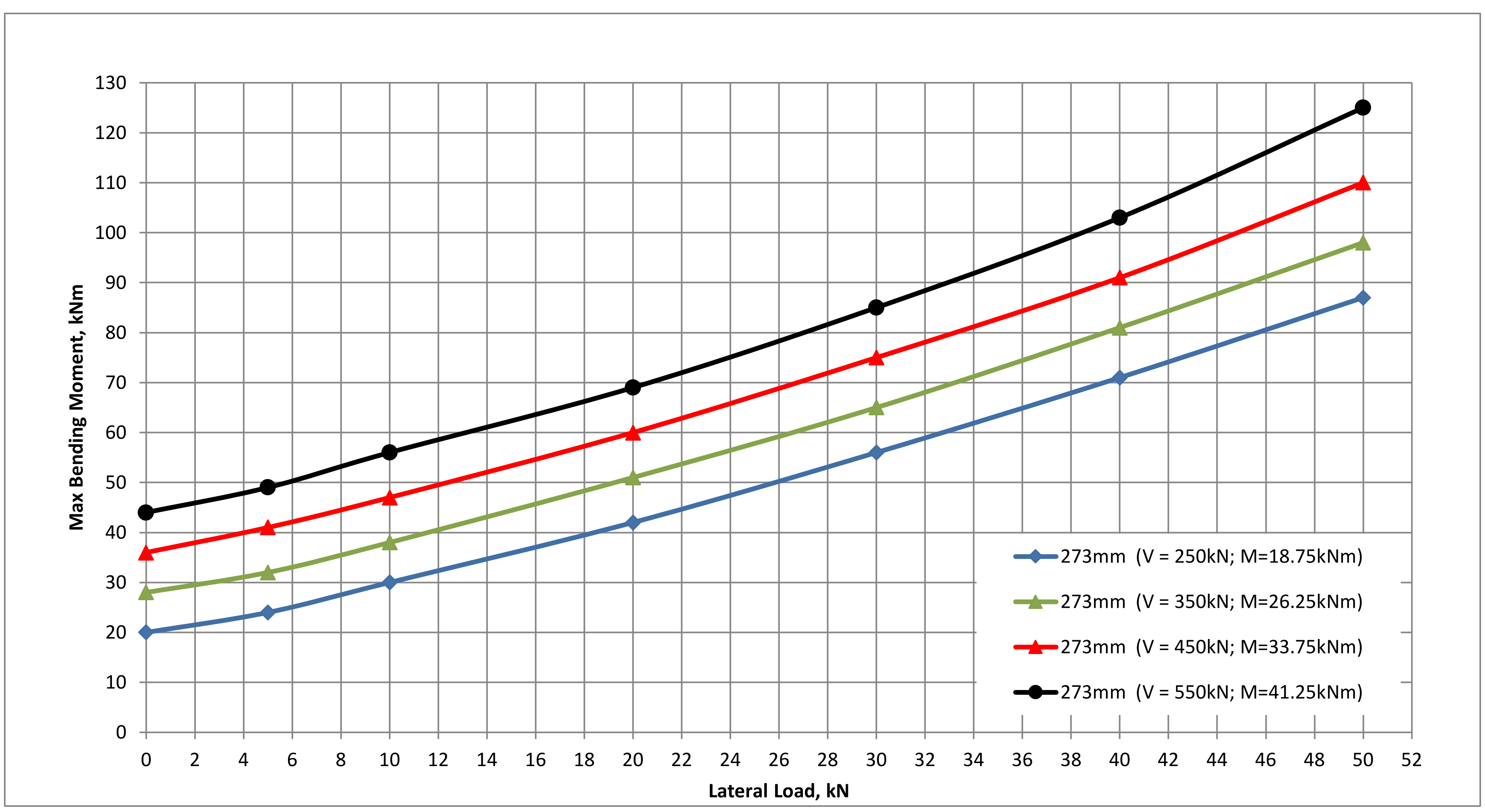
Project: SHRED
Pile Size: 273mm Dia. Pipe Pile

FREE-HEAD STEEL PILES - PILE HEAD AT 0.5m ABOVE FGL ASSUMED AT EL 661M (OR EL 2169 FT)

273mm (V = 250kN; M=18.75kNm)	Lateral Load (kN)	0	5	10	20	30	40	50
75mm eccentricity	Applied Moment (kNm)	18.75	18.75	18.75	18.75	18.75	18.75	18.75
	Deflection @ Grade (mm)	1.8	2.8	4.1	7.5	11.6	16.4	22.2
	Max BM (kNm)	20	24	30	42	56	71	87
273mm (V = 350kN; M=26.25kNm)	Lateral Load (kN)	0	5	10	20	30	40	50
75mm eccentricity	Applied Moment (kNm)	26.25	26.25	26.25	26.25	26.25	26.25	26.25
	Deflection @ Grade (mm)	2.7	3.9	5.5	9.2	13.6	18.9	25.4
	Max BM (kNm)	28	32	38	51	65	81	98
273mm (V = 450kN; M=33.75kNm)	Lateral Load (kN)	0	5	10	20	30	40	50
75mm eccentricity	Applied Moment (kNm)	33.75	33.75	33.75	33.75	33.75	33.75	33.75
	Deflection @ Grade (mm)	3.8	5.3	7.1	11.1	15.8	21.7	29.1
	Max BM (kNm)	36	41	47	60	75	91	110
273mm (V = 550kN; M=41.25kNm)	Lateral Load (kN)	0	5	10	20	30	40	50
75mm eccentricity	Applied Moment (kNm)	41.25	41.25	41.25	41.25	41.25	41.25	41.25
	Deflection @ Grade (mm)	5.1	6.8	8.8	13.1	18.4	25.0	35.7
	Max BM (kNm)	44	49	56	69	85	103	125

Note: Theoretical pile wall thickness is 9.5mm. 1.5mm corrosion allowance on wall thickness considered in analysis.



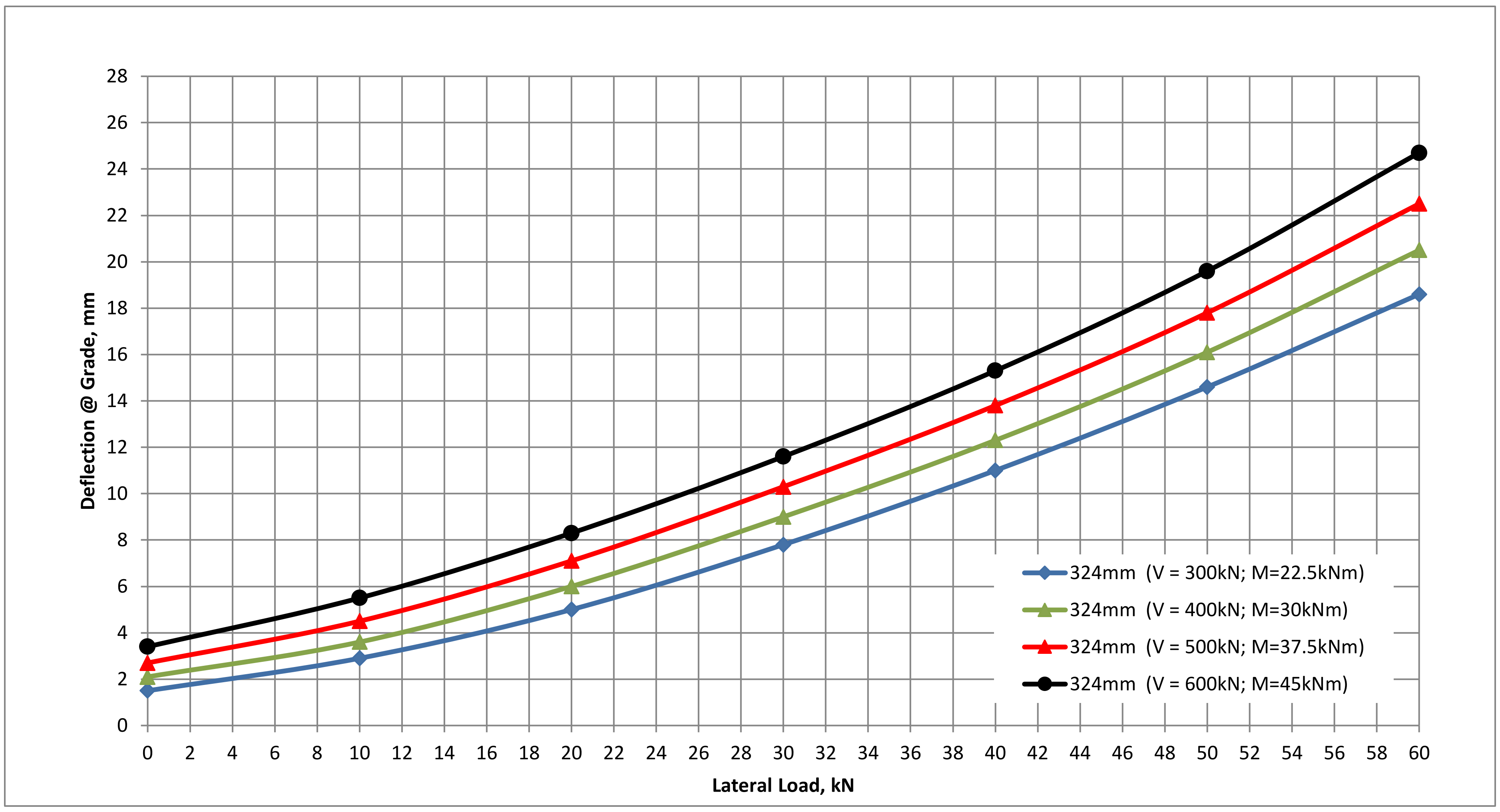


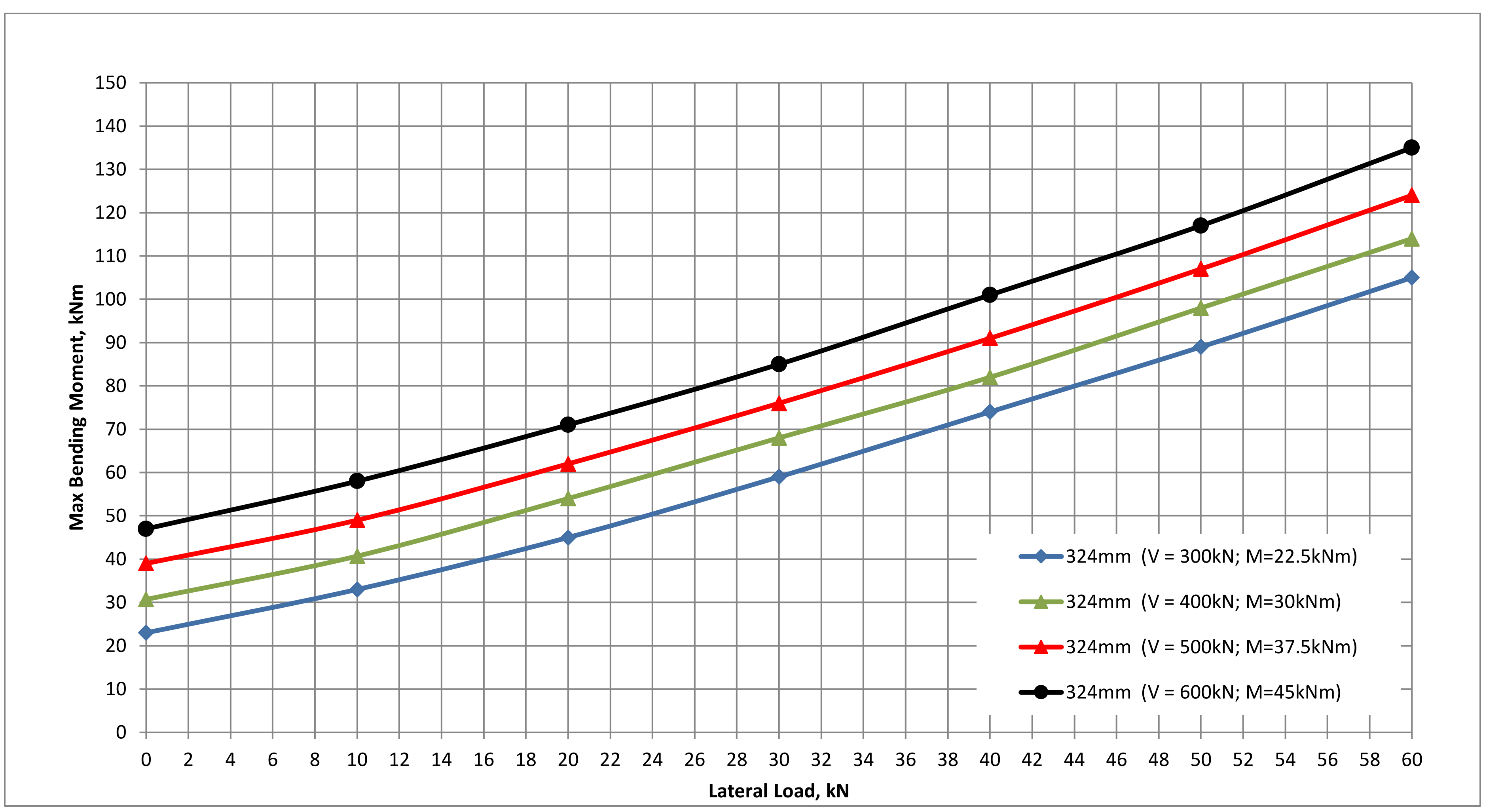
Project: SHRED
Pile Size: 324mm Dia. Pipe Pile

FREE-HEAD STEEL PILES - PILE HEAD AT 0.5m ABOVE FGL ASSUMED AT EL 661M (OR EL 2169 FT)

324mm (V = 300kN; M=22.5kNm)	Lateral Load (kN)	0	10	20	30	40	50	60
75mm eccentricity	Applied Moment (kNm)	22.5	22.5	22.5	22.5	22.5	22.5	22.5
	Deflection @ Grade (mm)	1.5	2.9	5.0	7.8	11.0	14.6	18.6
	Max BM (kNm)	23	33	45	59	74	89	105
324mm (V = 400kN; M=30kNm)	Lateral Load (kN)	0	10	20	30	40	50	60
75mm eccentricity	Applied Moment (kNm)	30	30	30	30	30	30	30
	Deflection @ Grade (mm)	2.1	3.6	6.0	9.0	12.3	16.1	20.5
	Max BM (kNm)	31	41	54	68	82	98	114
324mm (V = 500kN; M=37.5kNm)	Lateral Load (kN)	0	10	20	30	40	50	60
75mm eccentricity	Applied Moment (kNm)	37.5	37.5	37.5	37.5	37.5	37.5	37.5
	Deflection @ Grade (mm)	2.7	4.5	7.1	10.3	13.8	17.8	22.5
	Max BM (kNm)	39	49	62	76	91	107	124
324mm (V = 600kN; M=45kNm)	Lateral Load (kN)	0	10	20	30	40	50	60
75mm eccentricity	Applied Moment (kNm)	45	45	45	45	45	45	45
	Deflection @ Grade (mm)	3.4	5.5	8.3	11.6	15.3	19.6	24.7
	Max BM (kNm)	47	58	71	85	101	117	135

Note: Theoretical pile wall thickness is 9.5mm. 1.5mm corrosion allowance on wall thickness considered in analysis.



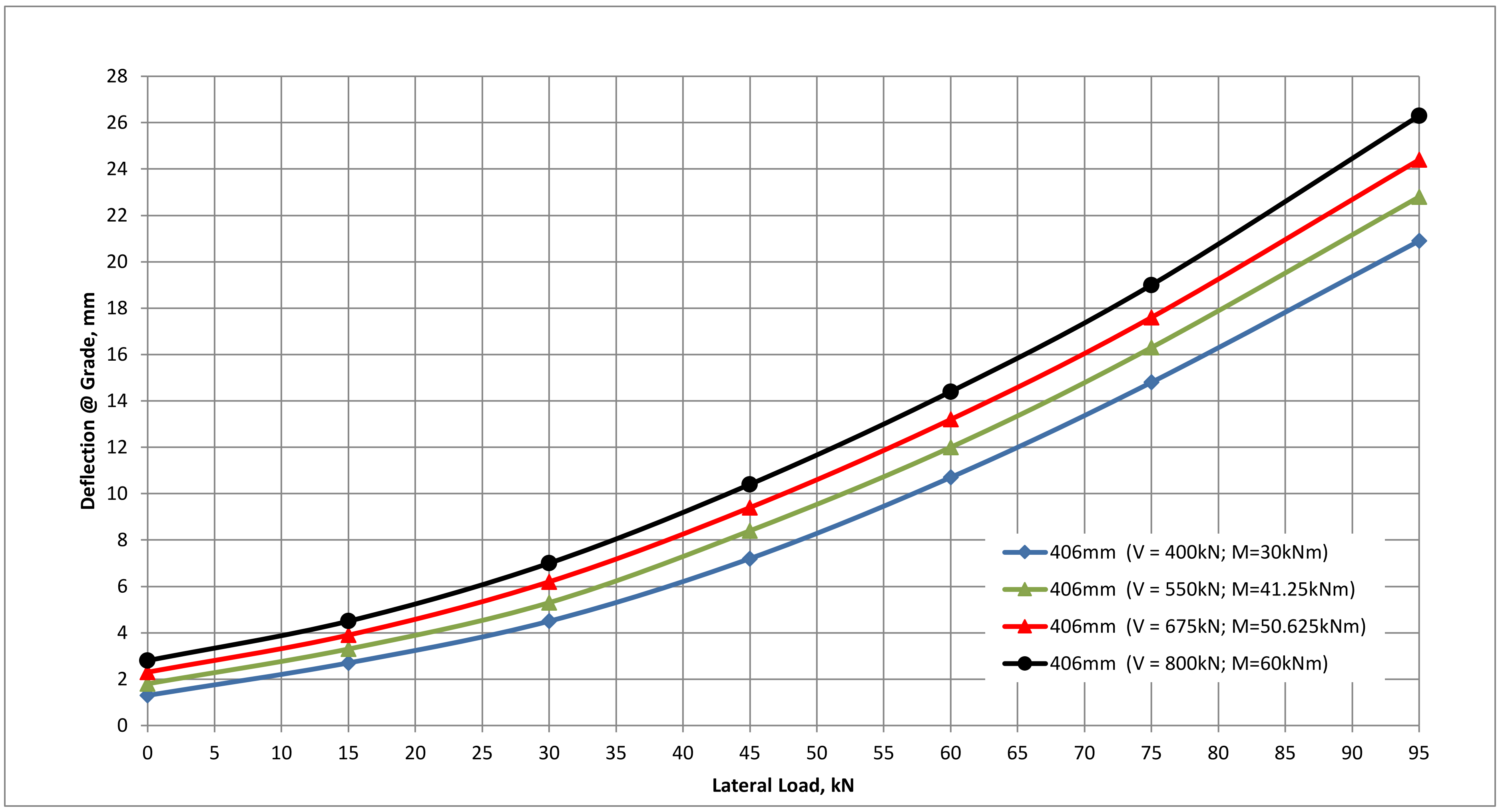


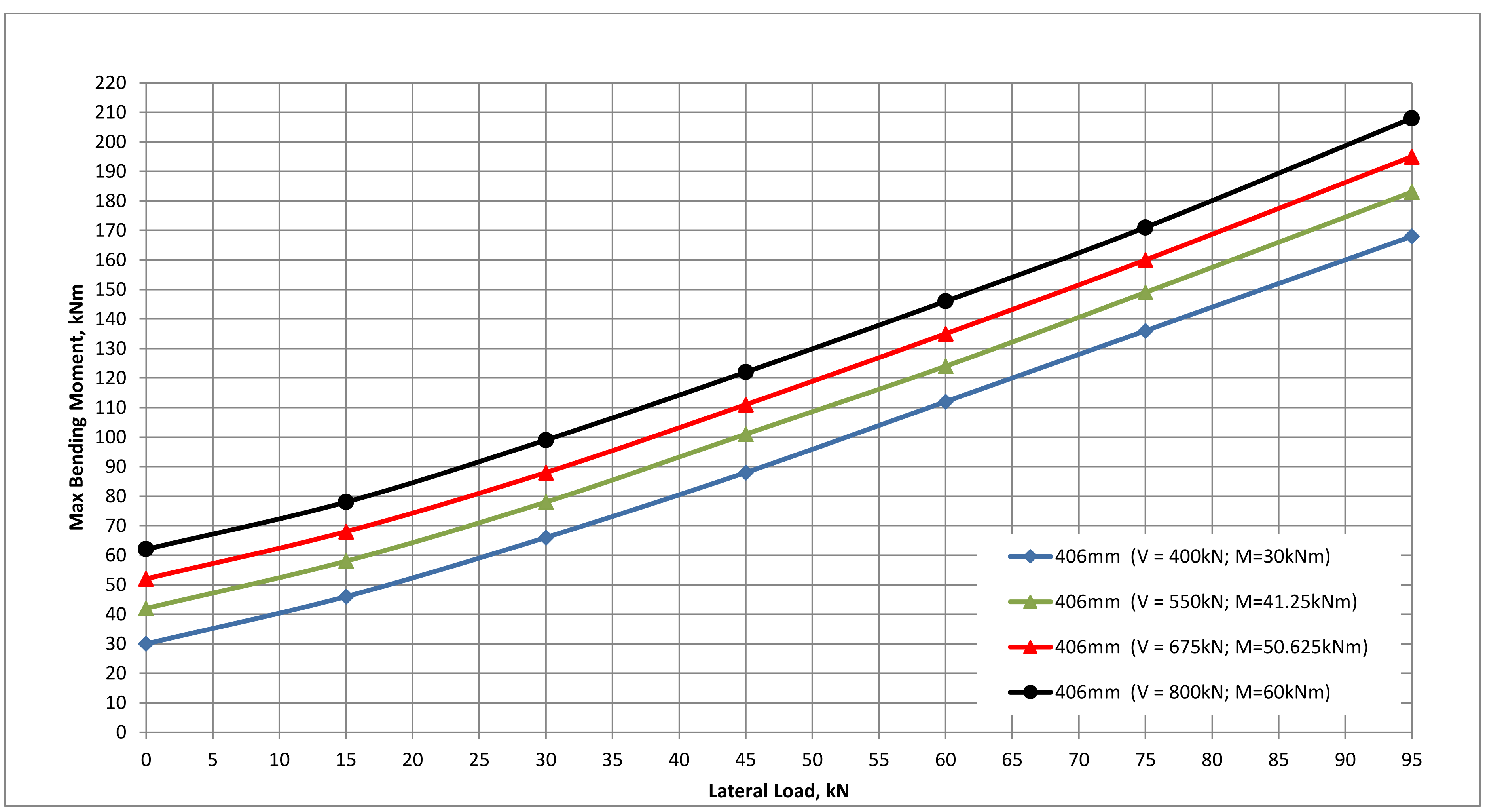
Project: SHRED
Pile Size: 406mm Dia. Pipe Pile

FREE-HEAD STEEL PILES - PILE HEAD AT 0.5m ABOVE FGL ASSUMED AT EL 661M (OR EL 2169 FT)

406mm (V = 400kN; M=30kNm)	Lateral Load (kN)	0	15	30	45	60	75	95
75mm eccentricity	Applied Moment (kNm)	30	30	30	30	30	30	30
	Deflection @ Grade (mm)	1.3	2.7	4.5	7.2	10.7	14.8	20.9
	Max BM (kNm)	30	46	66	88	112	136	168
406mm (V = 550kN; M=41.25kNm)	Lateral Load (kN)	0	15	30	45	60	75	95
75mm eccentricity	Applied Moment (kNm)	41.25	41.25	41.25	41.25	41.25	41.25	41.25
	Deflection @ Grade (mm)	1.8	3.3	5.3	8.4	12.0	16.3	22.8
	Max BM (kNm)	42	58	78	101	124	149	183
406mm (V = 675kN; M=50.625kNm)	Lateral Load (kN)	0	15	30	45	60	75	95
75mm eccentricity	Applied Moment (kNm)	50.63	50.63	50.63	50.63	50.63	50.63	50.63
	Deflection @ Grade (mm)	2.3	3.9	6.2	9.4	13.2	17.6	24.4
	Max BM (kNm)	52	68	88	111	135	160	195
406mm (V = 800kN; M=60kNm)	Lateral Load (kN)	0	15	30	45	60	75	95
75mm eccentricity	Applied Moment (kNm)	60	60	60	60	60	60	60
	Deflection @ Grade (mm)	2.8	4.5	7.0	10.4	14.4	19.0	26.3
	Max BM (kNm)	62	78	99	122	146	171	208

Note: Theoretical pile wall thickness is 9.5mm. 1.5mm corrosion allowance on wall thickness considered in analysis.



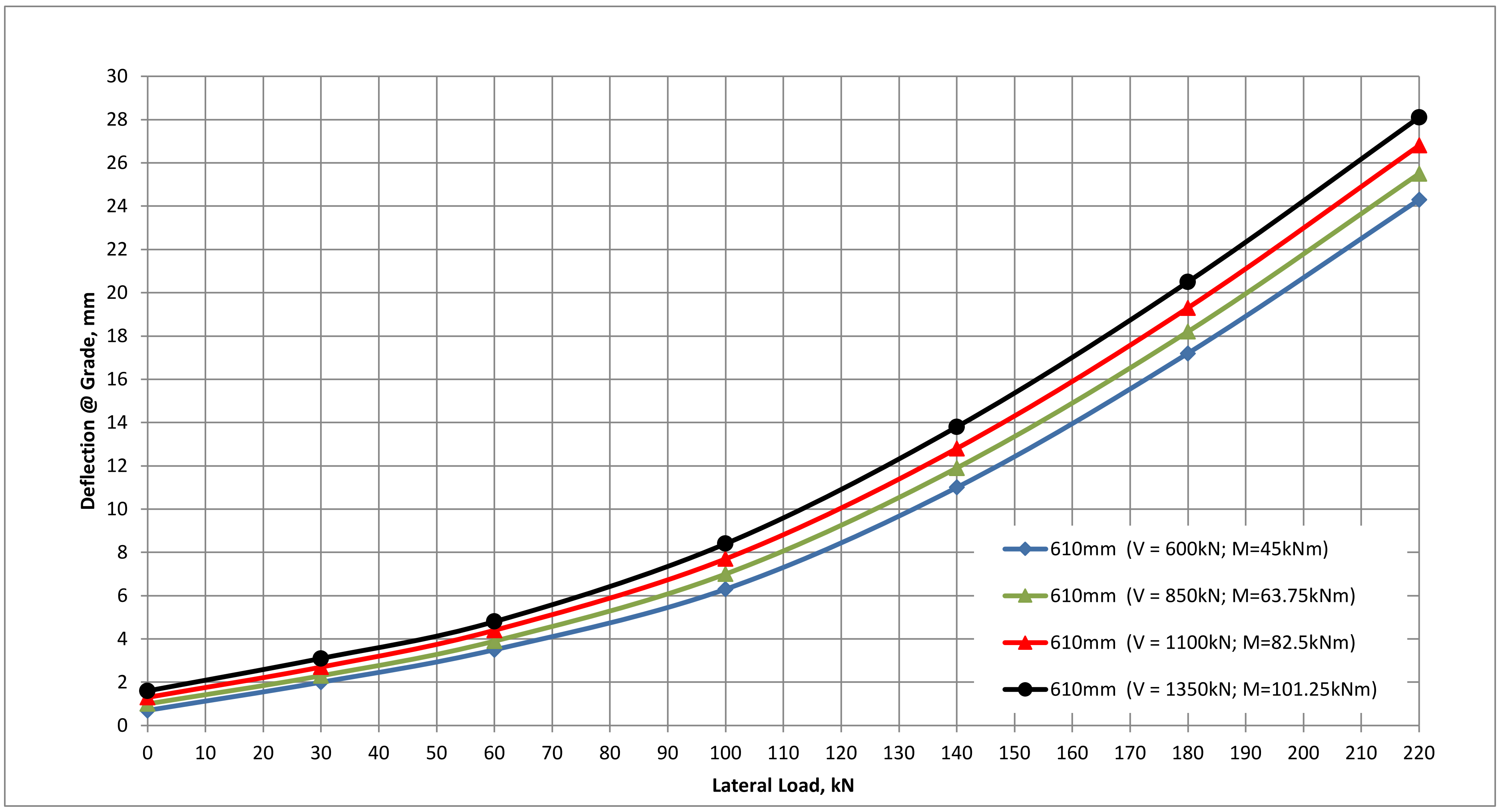


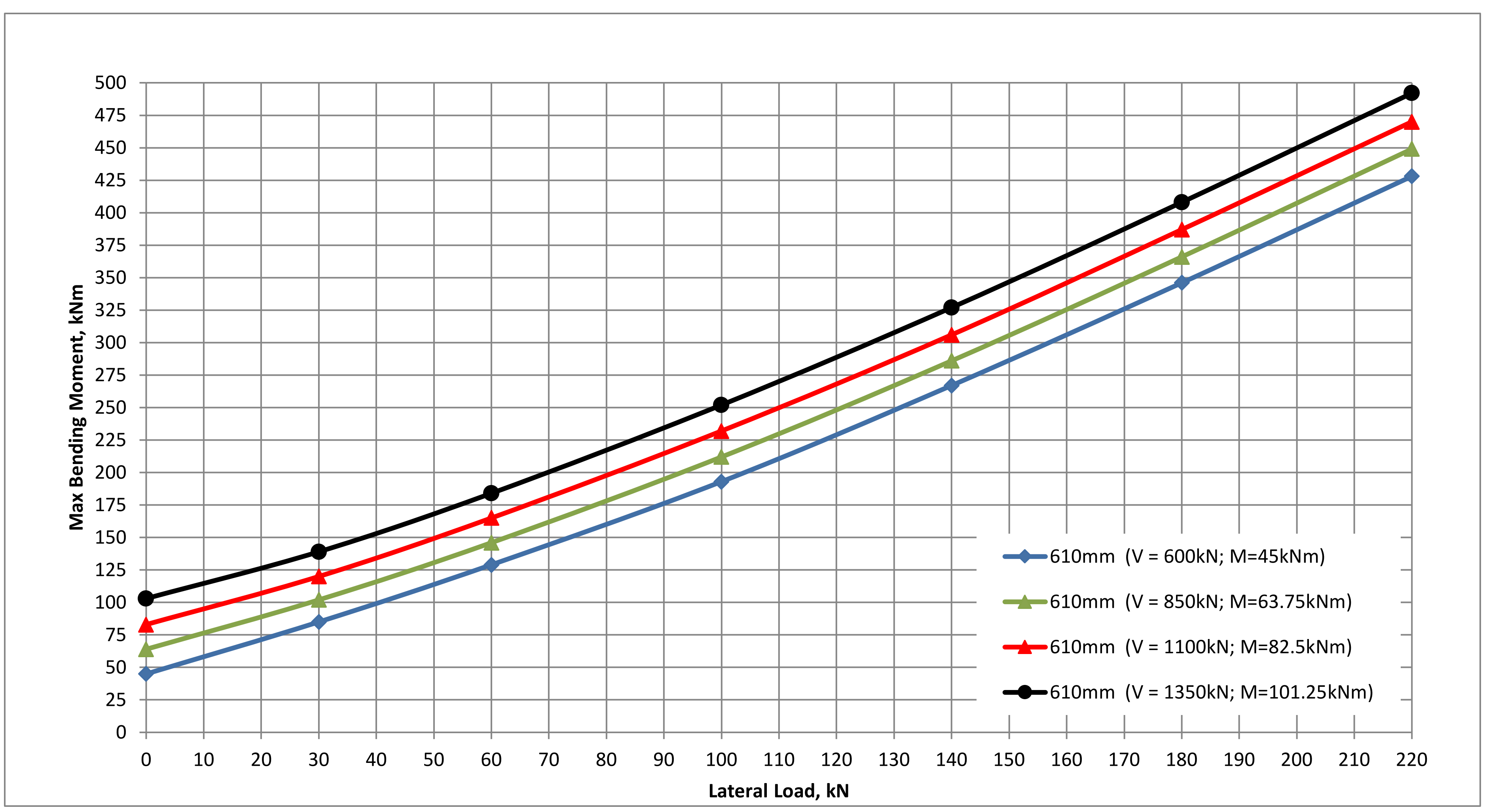
Project: SHRED
Pile Size: 610mm Dia. Pipe Pile

FREE-HEAD STEEL PILES - PILE HEAD AT 0.5m ABOVE FGL ASSUMED AT EL 661M (OR EL 2169 FT)

610mm (V = 600kN; M=45kNm)	Lateral Load (kN)	0	30	60	100	140	180	220
75mm eccentricity	Applied Moment (kNm)	45	45	45	45	45	45	45
	Deflection @ Grade (mm)	0.7	2.0	3.5	6.3	11.0	17.2	24.3
	Max BM (kNm)	45	85	129	193	267	346	428
610mm (V = 850kN; M=63.75kNm)	Lateral Load (kN)	0	30	60	100	140	180	220
75mm eccentricity	Applied Moment (kNm)	63.75	63.75	63.75	63.75	63.75	63.75	63.75
	Deflection @ Grade (mm)	1.0	2.3	3.9	7.0	11.9	18.2	25.5
	Max BM (kNm)	64	102	146	212	286	366	449
610mm (V = 1100kN; M=82.5kNm)	Lateral Load (kN)	0	30	60	100	140	180	220
75mm eccentricity	Applied Moment (kNm)	82.5	82.5	82.5	82.5	82.5	82.5	82.5
	Deflection @ Grade (mm)	1.3	2.7	4.4	7.7	12.8	19.3	26.8
	Max BM (kNm)	83	120	165	232	306	387	470
610mm (V = 1350kN; M=101.25kNm)	Lateral Load (kN)	0	30	60	100	140	180	220
75mm eccentricity	Applied Moment (kNm)	101.25	101.25	101.25	101.25	101.25	101.25	101.25
	Deflection @ Grade (mm)	1.6	3.1	4.8	8.4	13.8	20.5	28.1
	Max BM (kNm)	103	139	184	252	327	408	492

Note: Theoretical pile wall thickness is 12.7mm. 1.5mm corrosion allowance on wall thickness considered in analysis.



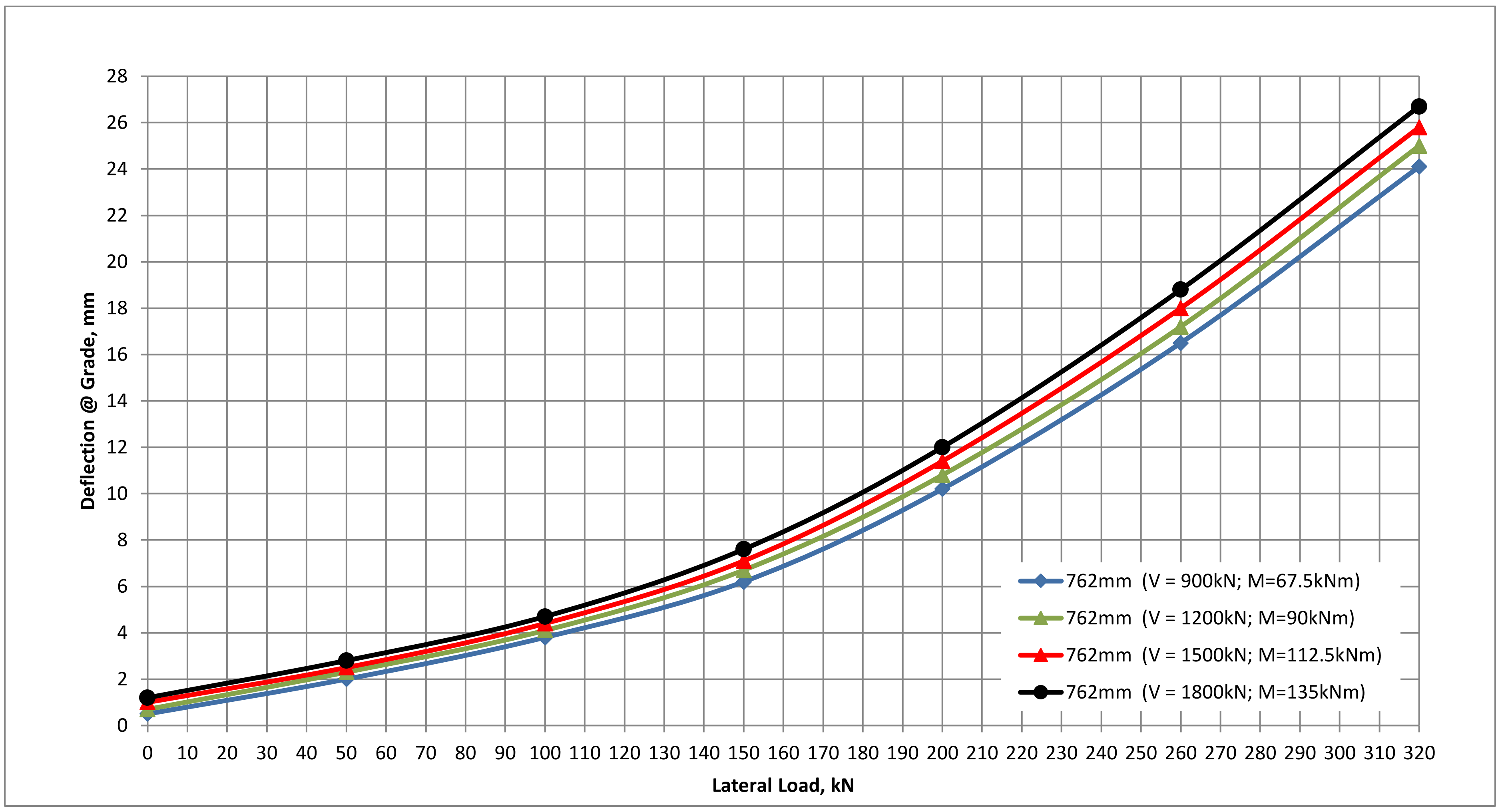


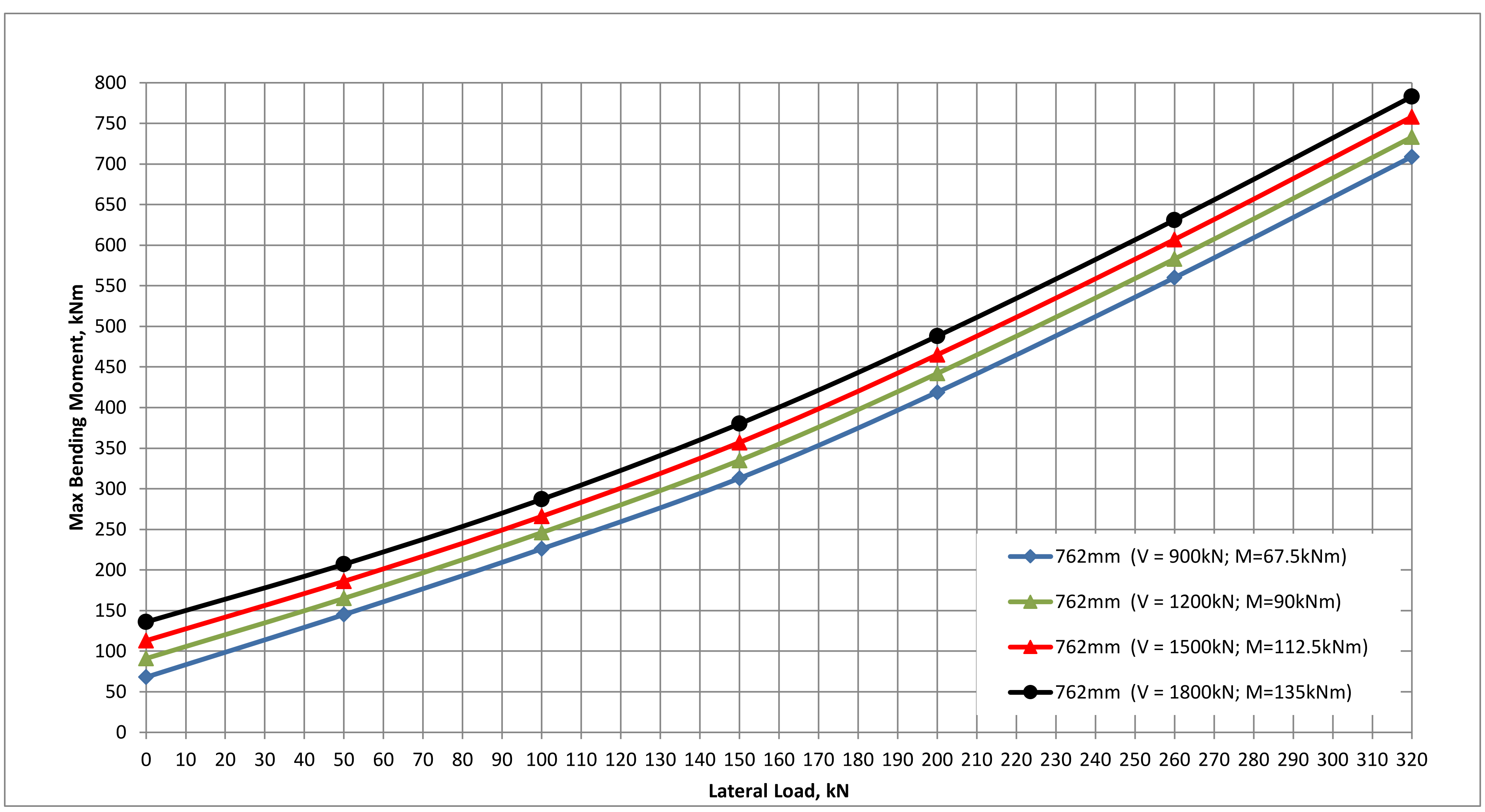
Project: SHRED
Pile Size: 762mm Dia. Pipe Pile

FREE-HEAD STEEL PILES - PILE HEAD AT 0.5m ABOVE FGL ASSUMED AT EL 661M (OR EL 2169 FT)

762mm (V = 900kN; M=67.5kNm)	Lateral Load (kN)	0	50	100	150	200	260	320
75mm eccentricity	Applied Moment (kNm)	67.5	67.5	67.5	67.5	67.5	67.5	67.5
	Deflection @ Grade (mm)	0.5	2.0	3.8	6.2	10.2	16.5	24.1
	Max BM (kNm)	68	145	226	313	419	560	709
762mm (V = 1200kN; M=90kNm)	Lateral Load (kN)	0	50	100	150	200	260	320
75mm eccentricity	Applied Moment (kNm)	90	90	90	90	90	90	90
	Deflection @ Grade (mm)	0.7	2.3	4.1	6.7	10.8	17.2	25.0
	Max BM (kNm)	91	165	246	335	442	583	733
762mm (V = 1500kN; M=112.5kNm)	Lateral Load (kN)	0	50	100	150	200	260	320
75mm eccentricity	Applied Moment (kNm)	112.5	112.5	112.5	112.5	112.5	112.5	112.5
	Deflection @ Grade (mm)	1.0	2.5	4.4	7.1	11.4	18.0	25.8
	Max BM (kNm)	113	186	266	357	465	607	758
762mm (V = 1800kN; M=135kNm)	Lateral Load (kN)	0	50	100	150	200	260	320
75mm eccentricity	Applied Moment (kNm)	135	135	135	135	135	135	135
	Deflection @ Grade (mm)	1.2	2.8	4.7	7.6	12.0	18.8	26.7
	Max BM (kNm)	136	207	287	380	488	631	783

Note: Theoretical pile wall thickness is 15.9mm. 1.5mm corrosion allowance on wall thickness considered in analysis.



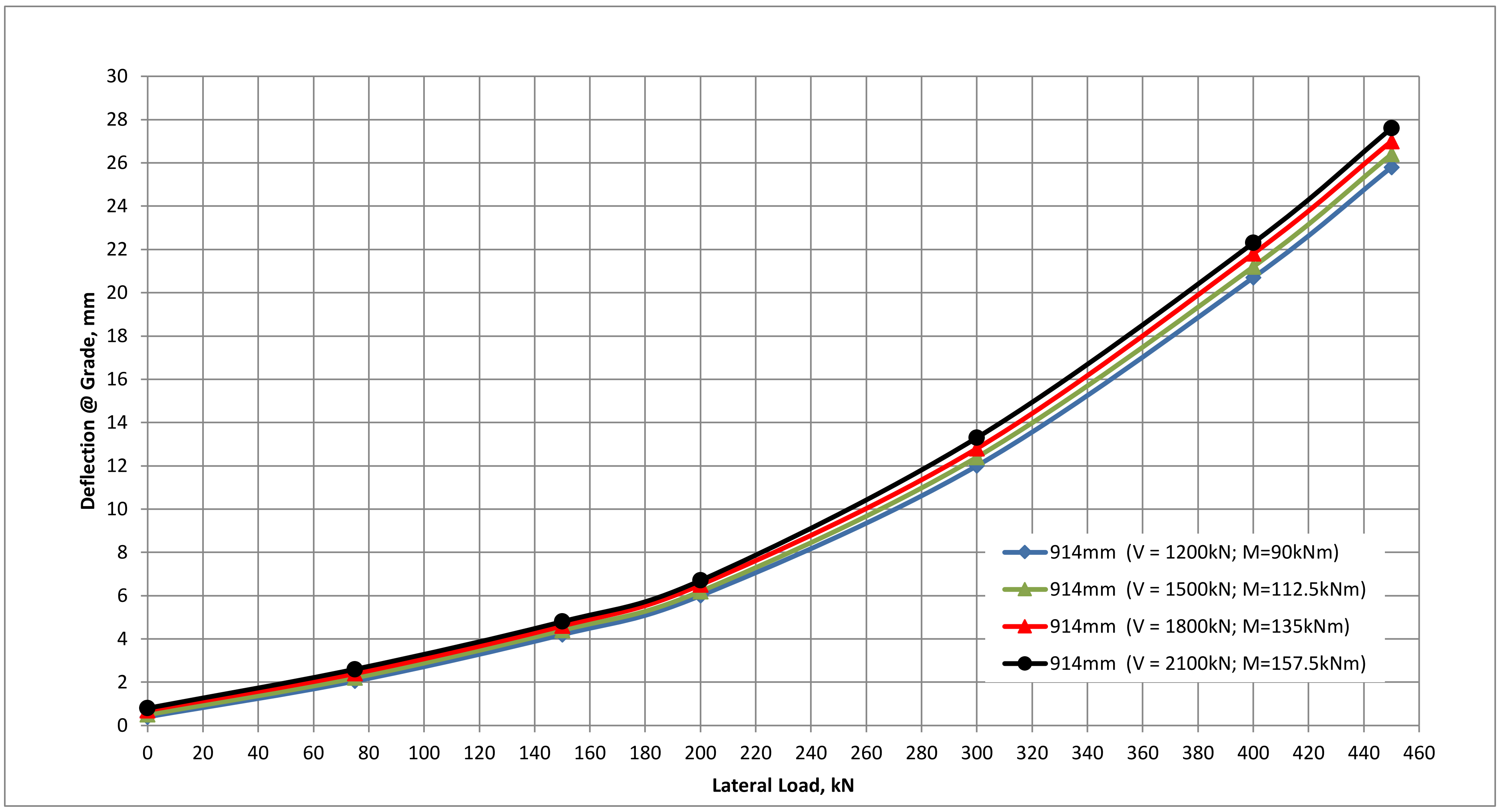


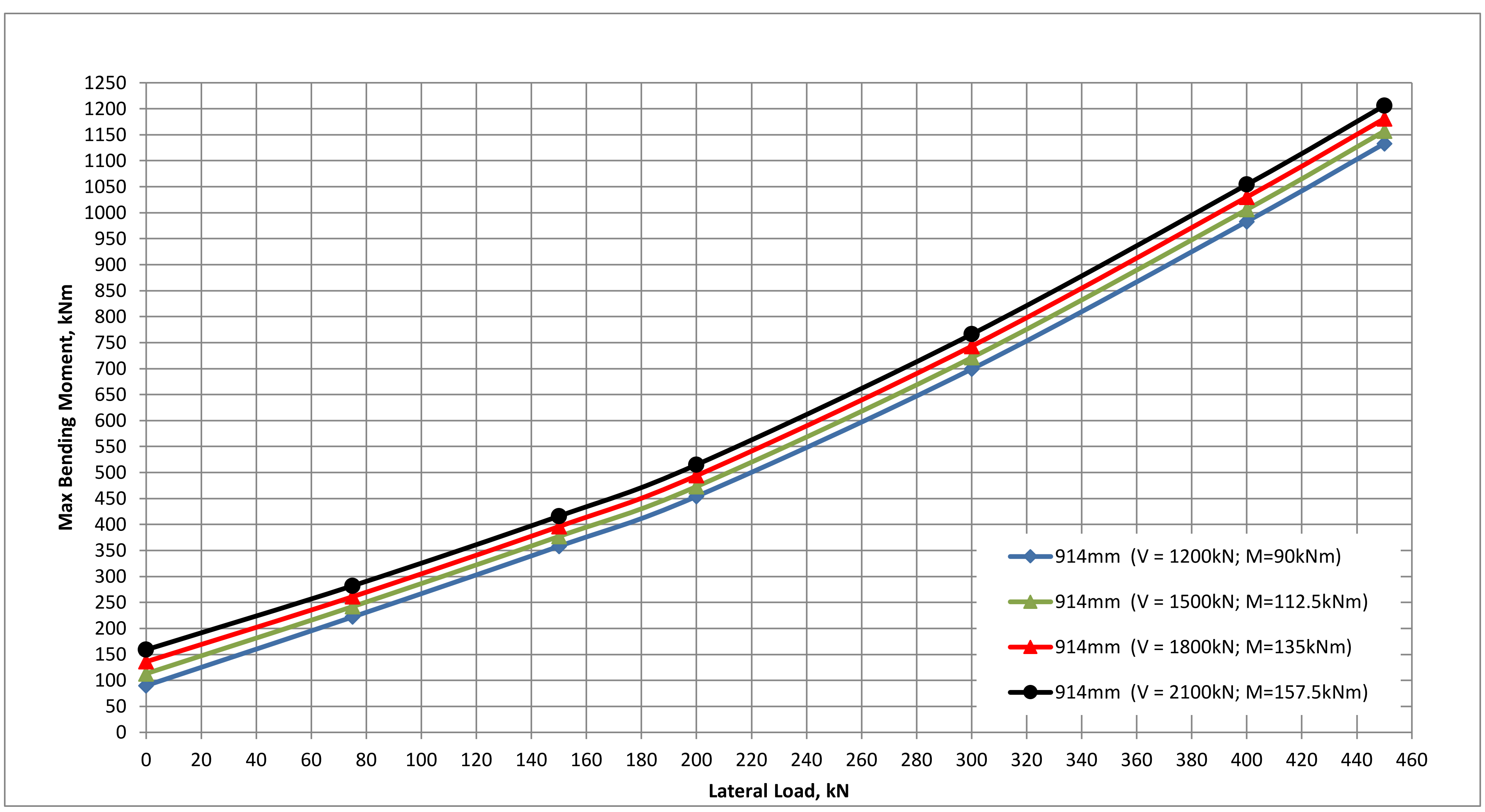
Project: SHRED
Pile Size: 914mm Dia. Pipe Pile

FREE-HEAD STEEL PILES - PILE HEAD AT 0.5m ABOVE FGL ASSUMED AT EL 661M (OR EL 2169 FT)

914mm (V = 1200kN; M=90kNm)	Lateral Load (kN)	0	75	150	200	300	400	450
75mm eccentricity	Applied Moment (kNm)	90	90	90	90	90	90	90
	Deflection @ Grade (mm)	0.4	2.1	4.2	6.0	12.0	20.7	25.8
	Max BM (kNm)	90	222	358	454	699	983	1133
914mm (V = 1500kN; M=112.5kNm)	Lateral Load (kN)	0	75	150	200	300	400	450
75mm eccentricity	Applied Moment (kNm)	112.5	112.5	112.5	112.5	112.5	112.5	112.5
	Deflection @ Grade (mm)	0.5	2.2	4.4	6.2	12.4	21.2	26.4
	Max BM (kNm)	113	242	377	473	721	1006	1157
914mm (V = 1800kN; M=135kNm)	Lateral Load (kN)	0	75	150	200	300	400	450
75mm eccentricity	Applied Moment (kNm)	135	135	135	135	135	135	135
	Deflection @ Grade (mm)	0.7	2.4	4.6	6.5	12.8	21.8	27.0
	Max BM (kNm)	136	261	396	494	743	1030	1181
914mm (V = 2100kN; M=157.5kNm)	Lateral Load (kN)	0	75	150	200	300	400	450
75mm eccentricity	Applied Moment (kNm)	157.5	157.5	157.5	157.5	157.5	157.5	157.5
	Deflection @ Grade (mm)	0.8	2.6	4.8	6.7	13.3	22.3	27.6
	Max BM (kNm)	159	282	416	515	766	1054	1206

Note: Theoretical pile wall thickness is 19.0mm. 1.5mm corrosion allowance on wall thickness considered in analysis.





DEEP FOUNDATION PHILOSOPHY

APPENDIX C

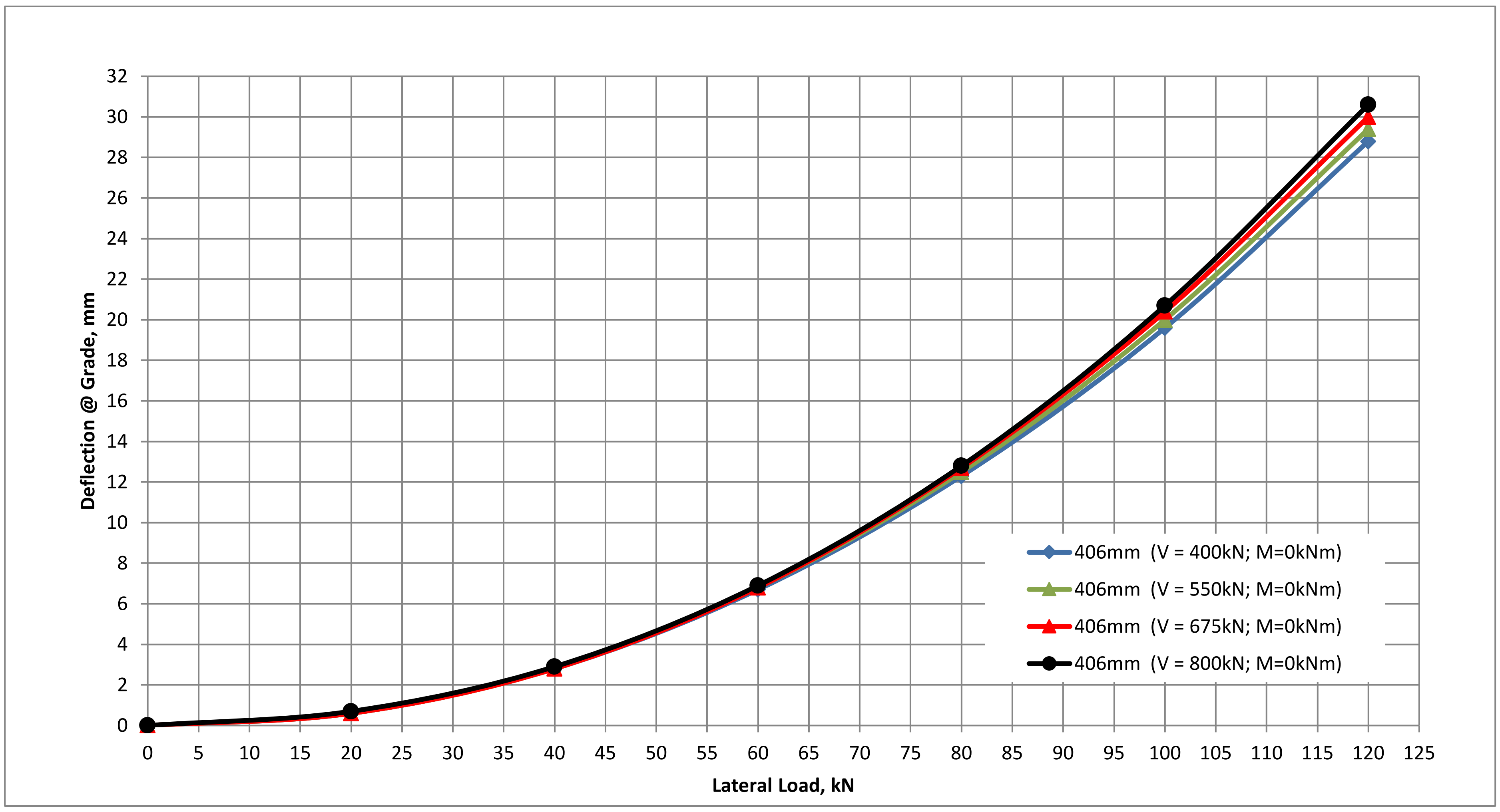
Lateral Analyses of Single Piles (free head)
Pile Head at 2.0m below Finished Grade

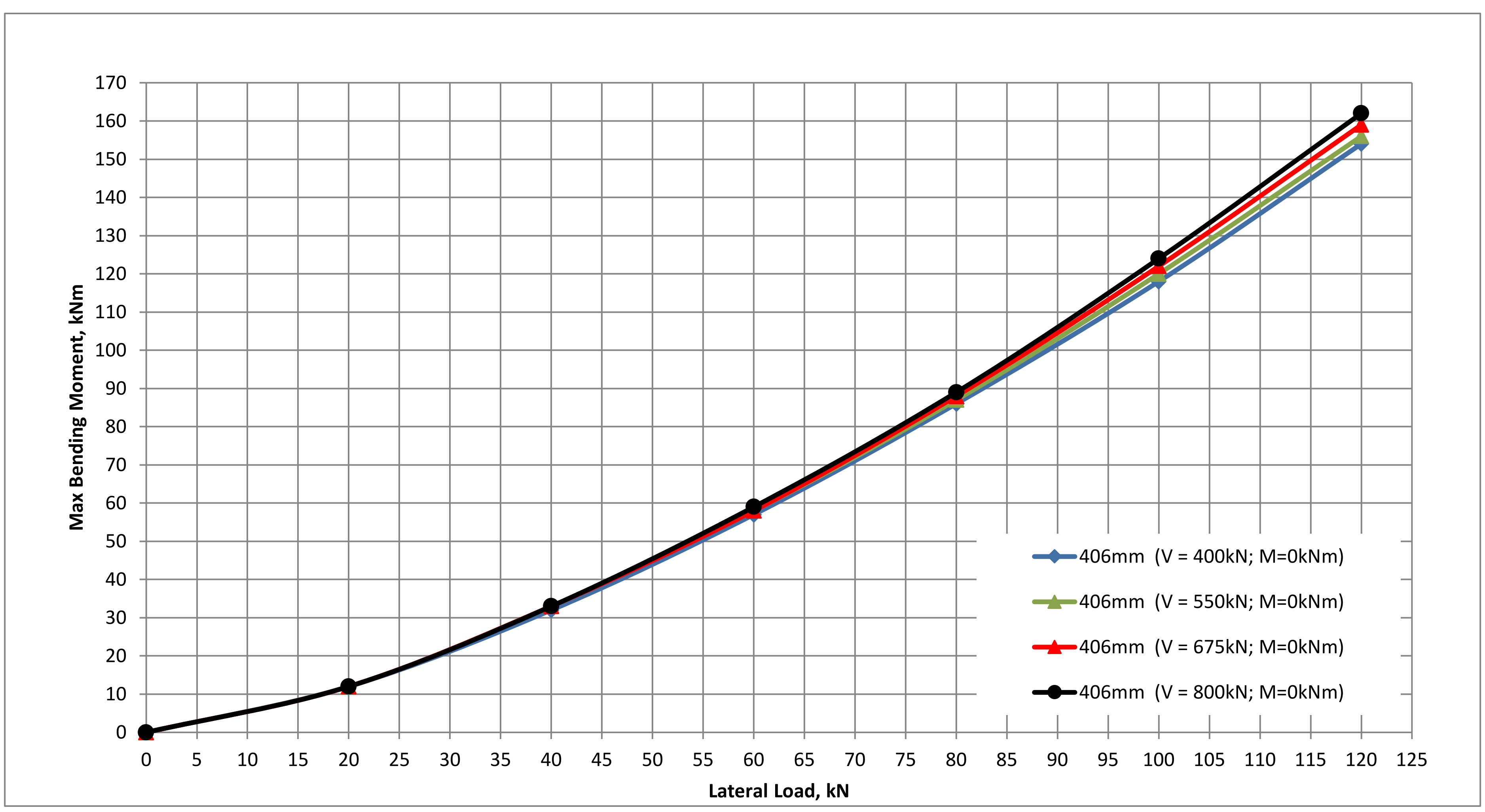
Project: SHRED
Pile Size: 406mm Dia. Pipe Pile

FREE-HEAD STEEL PILES - PILE HEAD AT EL 2164 FT (or EL 659.5M)

406mm (V = 400kN; M=0kNm)	Lateral Load (kN)	0	20	40	60	80	100	120
	Applied Moment (kNm)	0	0	0	0	0	0	0
	Deflection @ Grade (mm)	0	0.6	2.8	6.7	12.3	19.6	28.8
	Max BM (kNm)	0	12	32	57	86	118	154
406mm (V = 550kN; M=0kNm)	Lateral Load (kN)	0	20	40	60	80	100	120
	Applied Moment (kNm)	0	0	0	0	0	0	0
	Deflection @ Grade (mm)	0	0.6	2.8	6.8	12.5	20.0	29.4
	Max BM (kNm)	0	12	33	58	87	120	156
406mm (V = 675kN; M=0kNm)	Lateral Load (kN)	0	20	40	60	80	100	120
	Applied Moment (kNm)	0	0	0	0	0	0	0
	Deflection @ Grade (mm)	0	0.6	2.8	6.8	12.7	20.4	30.0
	Max BM (kNm)	0	12	33	58	88	122	159
406mm (V = 800kN; M=0kNm)	Lateral Load (kN)	0	20	40	60	80	100	120
	Applied Moment (kNm)	0	0	0	0	0	0	0
	Deflection @ Grade (mm)	0	0.7	2.9	6.9	12.8	20.7	30.6
	Max BM (kNm)	0	12	33	59	89	124	162

Note: Theoretical pile wall thickness is 9.5mm. 1.5mm corrosion allowance on wall thickness considered in analysis.



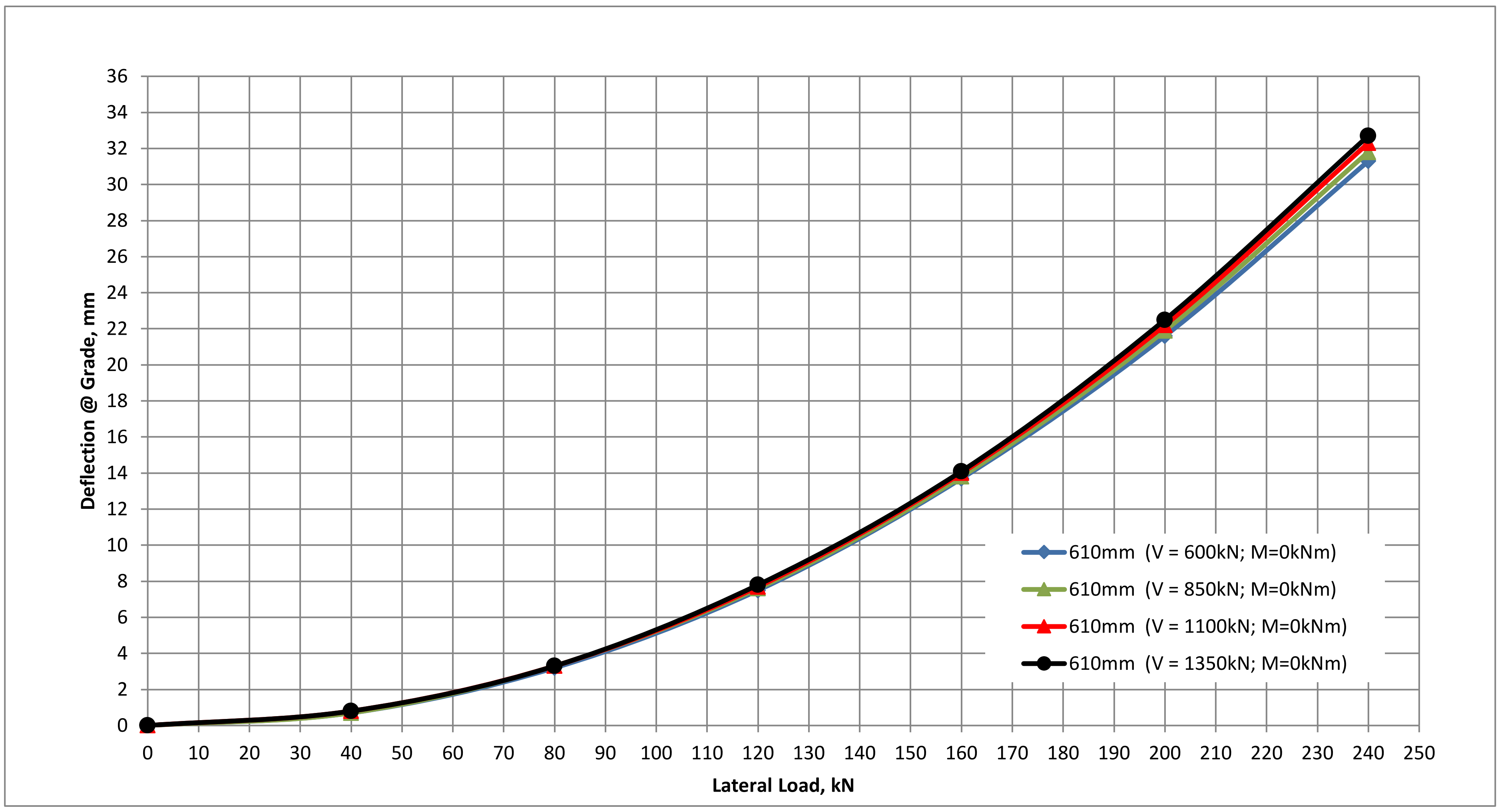


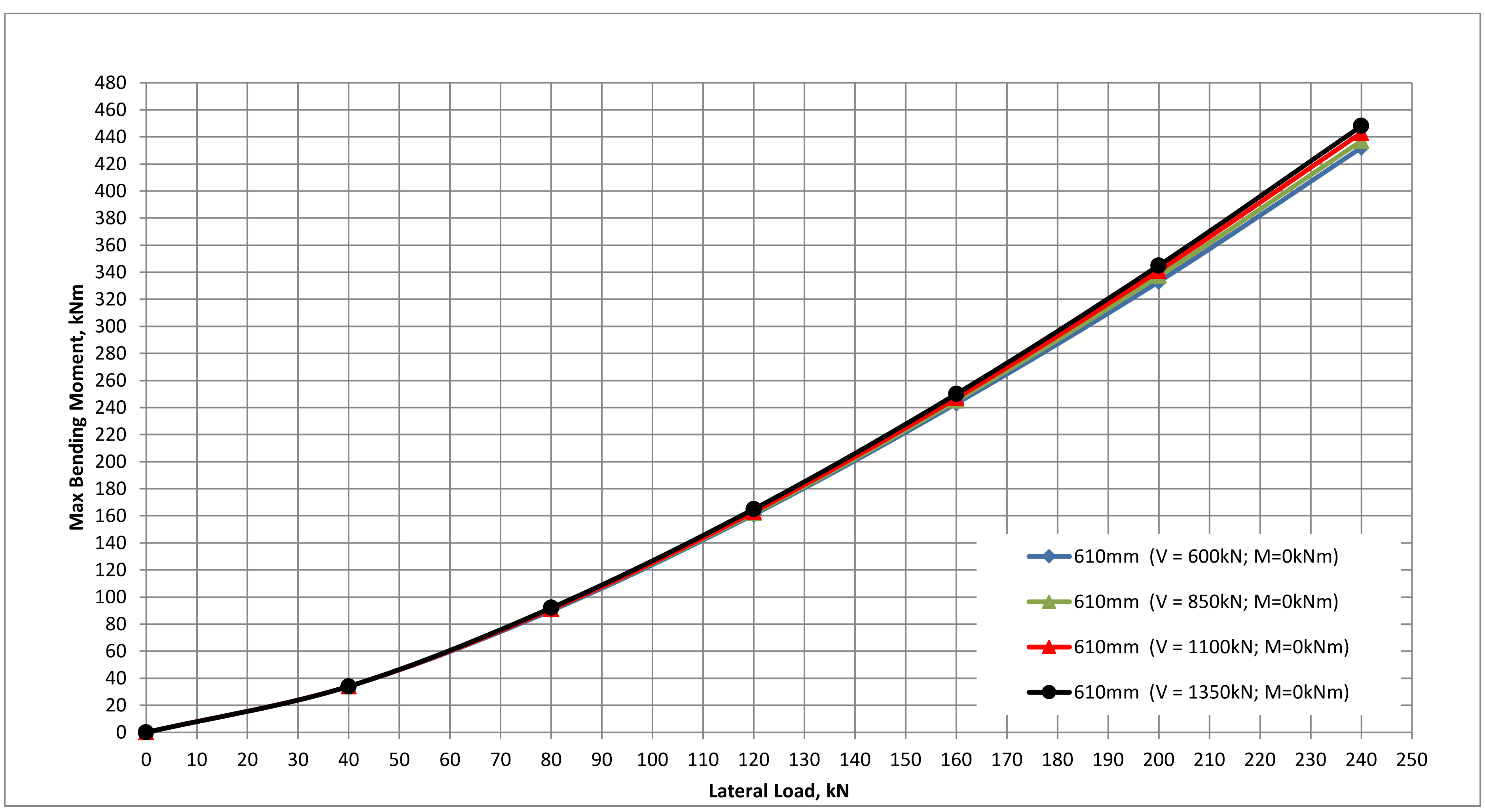
Project: SHRED
Pile Size: 610mm Dia. Pipe Pile

FREE-HEAD STEEL PILES - PILE HEAD AT EL 2164 FT (or EL 659.5M)

610mm (V = 600kN; M=0kNm)	Lateral Load (kN)	0	40	80	120	160	200	240
	Applied Moment (kNm)	0	0	0	0	0	0	0
	Deflection @ Grade (mm)	0	0.7	3.2	7.5	13.7	21.6	31.3
	Max BM (kNm)	0	34	90	161	243	333	432
610mm (V = 850kN; M=0kNm)	Lateral Load (kN)	0	40	80	120	160	200	240
	Applied Moment (kNm)	0	0	0	0	0	0	0
	Deflection @ Grade (mm)	0	0.7	3.3	7.6	13.8	21.9	31.8
	Max BM (kNm)	0	34	91	162	245	337	437
610mm (V = 1100kN; M=0kNm)	Lateral Load (kN)	0	40	80	120	160	200	240
	Applied Moment (kNm)	0	0	0	0	0	0	0
	Deflection @ Grade (mm)	0	0.8	3.3	7.7	14.0	22.2	32.3
	Max BM (kNm)	0	34	91	163	247	341	443
610mm (V = 1350kN; M=0kNm)	Lateral Load (kN)	0	40	80	120	160	200	240
	Applied Moment (kNm)	0	0	0	0	0	0	0
	Deflection @ Grade (mm)	0	0.8	3.3	7.8	14.1	22.5	32.7
	Max BM (kNm)	0	34	92	165	250	345	448

Note: Theoretical pile wall thickness is 12.7mm. 1.5mm corrosion allowance on wall thickness considered in analysis.



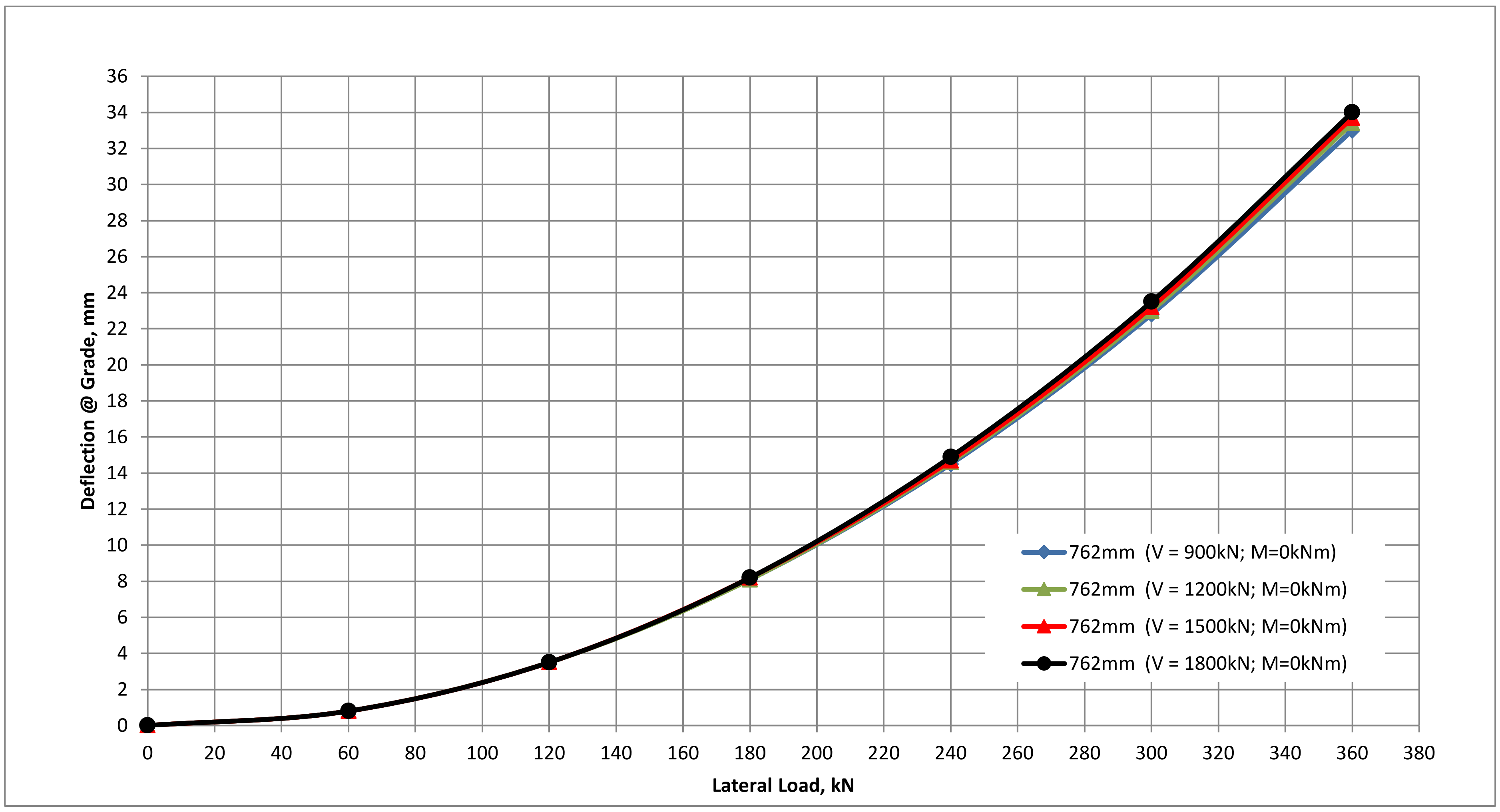


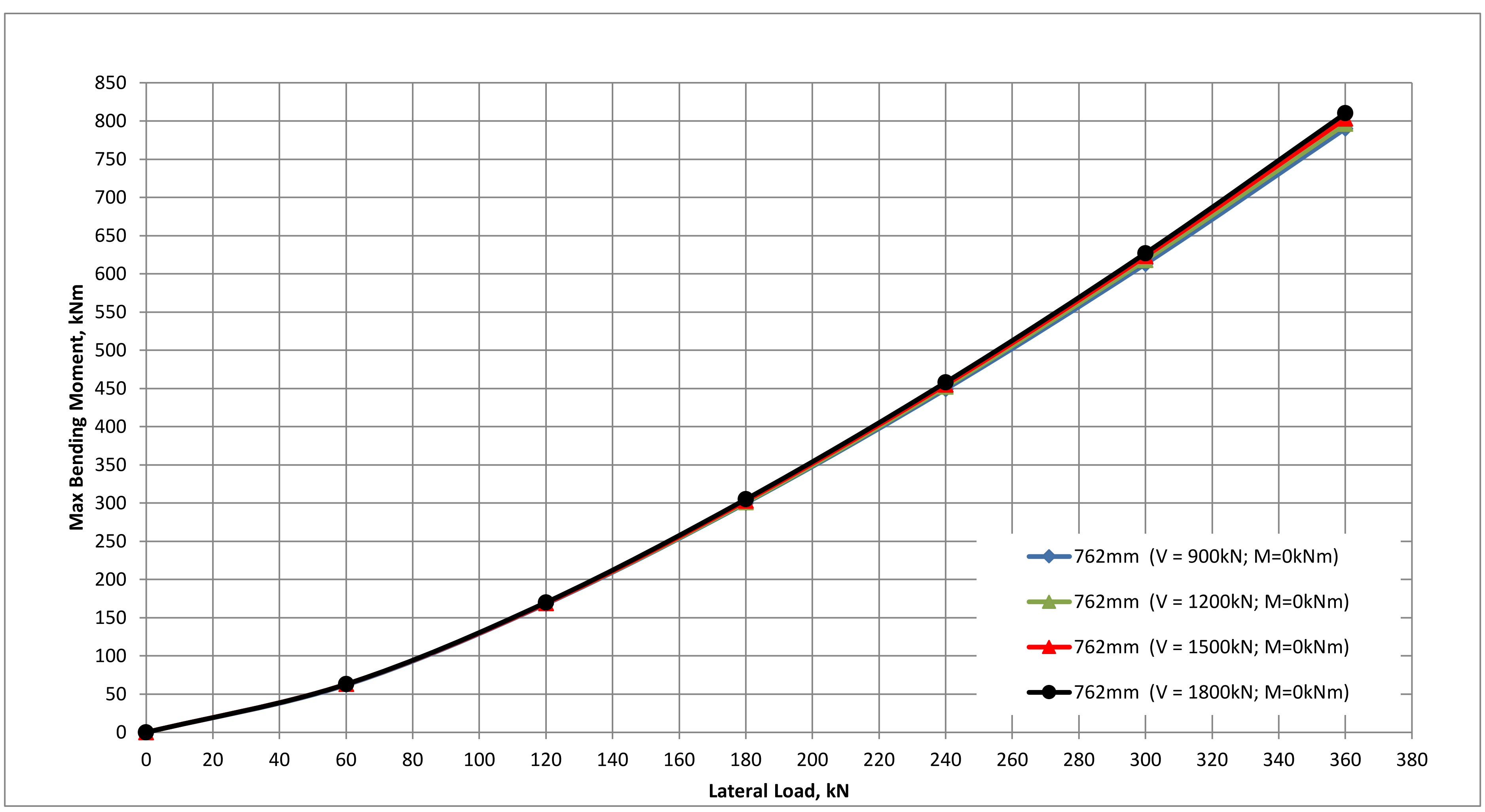
Project: SHRED
Pile Size: 762mm Dia. Pipe Pile

FREE-HEAD STEEL PILES - PILE HEAD AT EL 2164 FT (or EL 659.5M)

762mm (V = 900kN; M=0kNm)	Lateral Load (kN)	0	60	120	180	240	300	360
	Applied Moment (kNm)	0	0	0	0	0	0	0
	Deflection @ Grade (mm)	0	0.8	3.5	8.1	14.5	22.8	33.0
	Max BM (kNm)	0	62	168	300	449	613	790
762mm (V = 1200kN; M=0kNm)	Lateral Load (kN)	0	60	120	180	240	300	360
	Applied Moment (kNm)	0	0	0	0	0	0	0
	Deflection @ Grade (mm)	0	0.8	3.5	8.1	14.6	23.0	33.4
	Max BM (kNm)	0	63	169	301	452	618	796
762mm (V = 1500kN; M=0kNm)	Lateral Load (kN)	0	60	120	180	240	300	360
	Applied Moment (kNm)	0	0	0	0	0	0	0
	Deflection @ Grade (mm)	0	0.8	3.5	8.2	14.7	23.2	33.7
	Max BM (kNm)	0	63	169	303	455	623	803
762mm (V = 1800kN; M=0kNm)	Lateral Load (kN)	0	60	120	180	240	300	360
	Applied Moment (kNm)	0	0	0	0	0	0	0
	Deflection @ Grade (mm)	0	0.8	3.5	8.2	14.9	23.5	34.0
	Max BM (kNm)	0	63	170	305	458	627	810

Note: Theoretical pile wall thickness is 15.9mm. 1.5mm corrosion allowance on wall thickness considered in analysis.



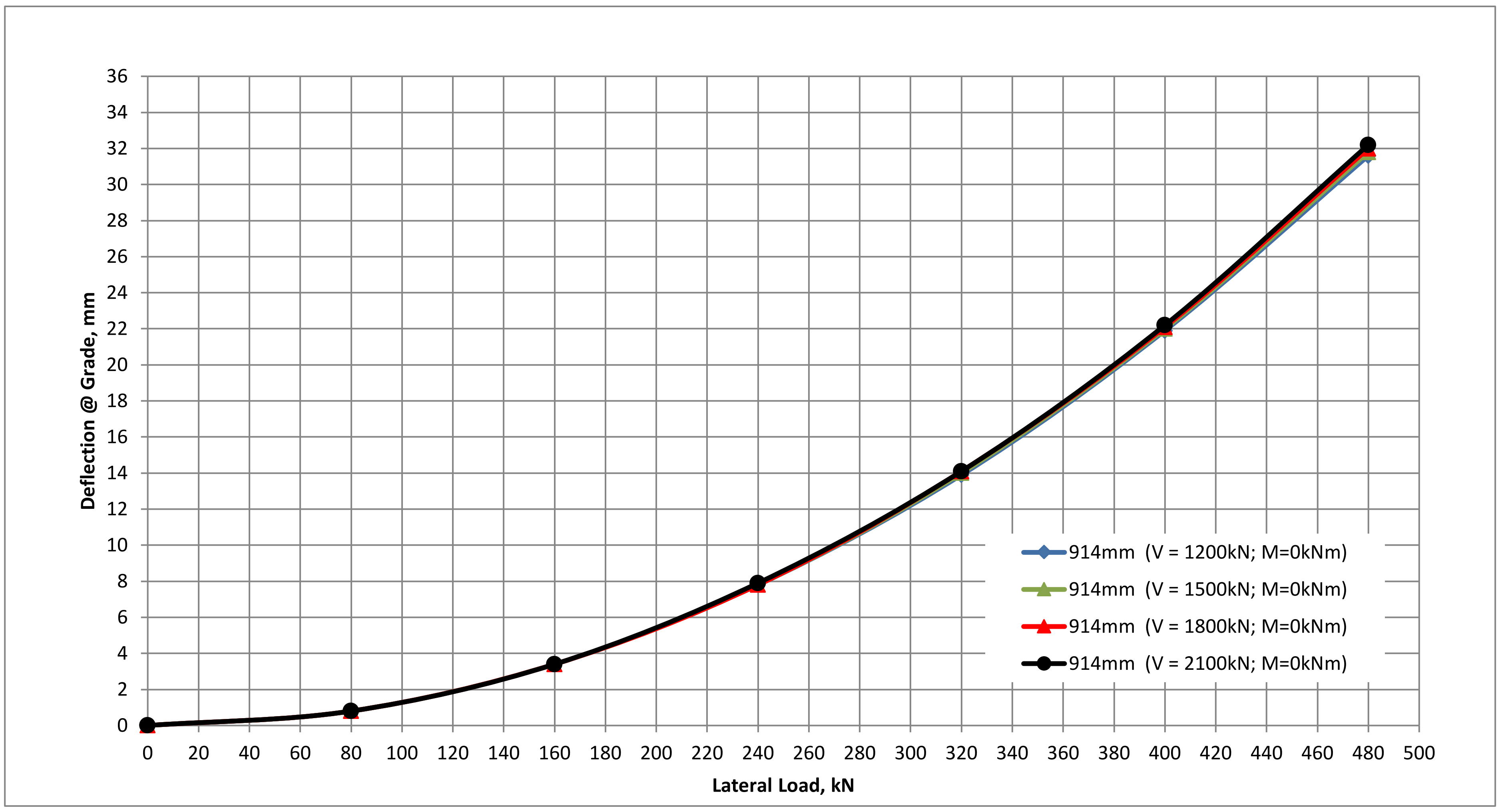


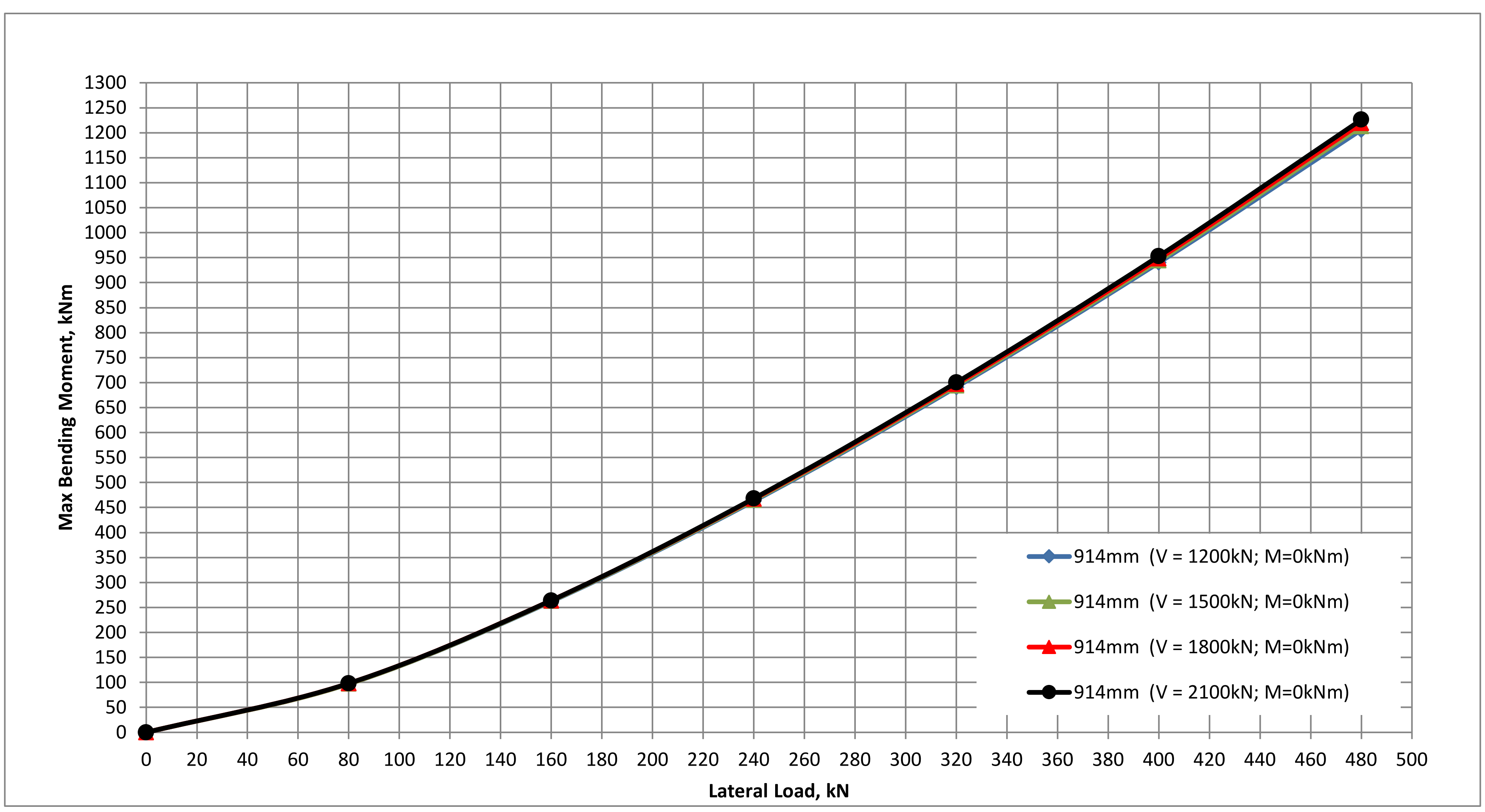
Project: SHRED
Pile Size: 914mm Dia. Pipe Pile

FREE-HEAD STEEL PILES - PILE HEAD AT EL 2164 FT (or EL 659.5M)

914mm (V = 1200kN; M=0kNm)	Lateral Load (kN)	0	80	160	240	320	400	480
	Applied Moment (kNm)	0	0	0	0	0	0	0
	Deflection @ Grade (mm)	0	0.8	3.4	7.8	13.9	21.9	31.6
	Max BM (kNm)	0	97	262	463	691	940	1206
914mm (V = 1500kN; M=0kNm)	Lateral Load (kN)	0	80	160	240	320	400	480
	Applied Moment (kNm)	0	0	0	0	0	0	0
	Deflection @ Grade (mm)	0	0.8	3.4	7.8	14.0	22.0	31.8
	Max BM (kNm)	0	97	263	465	694	944	1213
914mm (V = 1800kN; M=0kNm)	Lateral Load (kN)	0	80	160	240	320	400	480
	Applied Moment (kNm)	0	0	0	0	0	0	0
	Deflection @ Grade (mm)	0	0.8	3.4	7.8	14.1	22.1	32.0
	Max BM (kNm)	0	98	264	467	697	948	1219
914mm (V = 2100kN; M=0kNm)	Lateral Load (kN)	0	80	160	240	320	400	480
	Applied Moment (kNm)	0	0	0	0	0	0	0
	Deflection @ Grade (mm)	0	0.8	3.4	7.9	14.1	22.2	32.2
	Max BM (kNm)	0	98	264	468	700	953	1226

Note: Theoretical pile wall thickness is 19.0mm. 1.5mm corrosion allowance on wall thickness considered in analysis.





DEEP FOUNDATION PHILOSOPHY

APPENDIX D

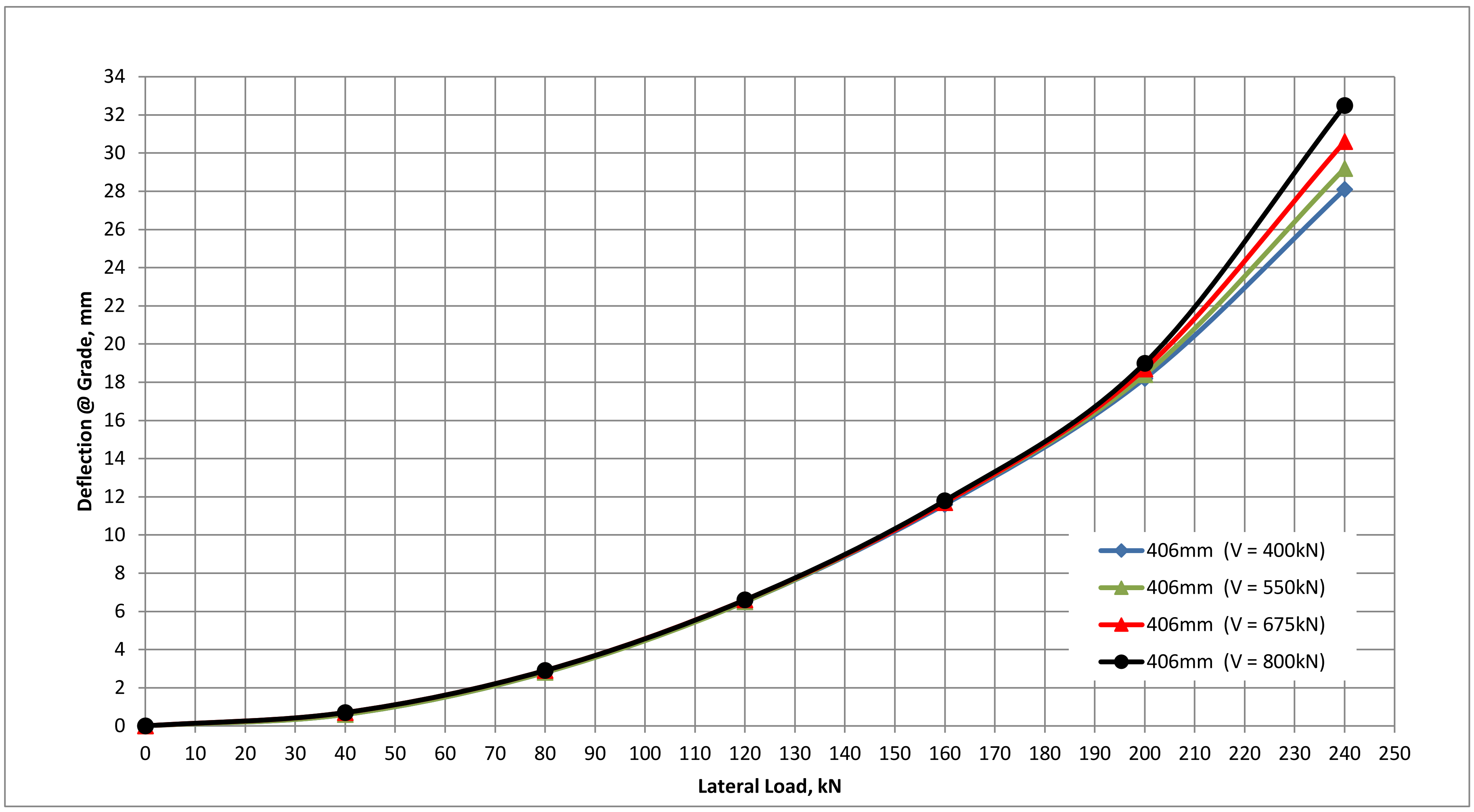
Lateral Analyses of Single Piles (fixed head)
Pile Head at 2.0m below Finished Grade

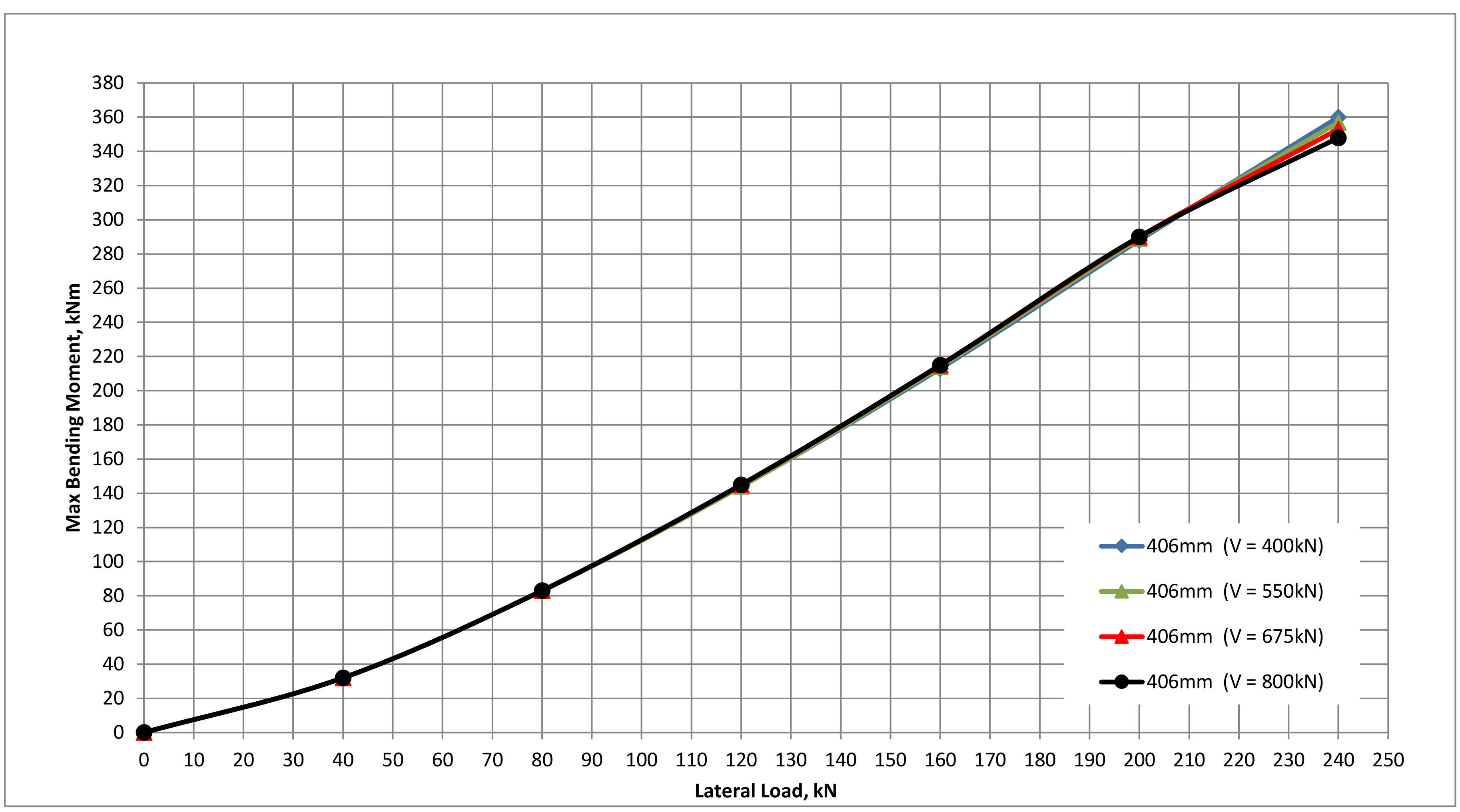
Project: SHRED
Pile Size: 406mm Dia. Pipe Pile

FIXED-HEAD STEEL PILES - PILE HEAD AT EL 2164 FT (or EL 659.5M)

406mm (V = 400kN)	Lateral Load (kN)	0	40	80	120	160	200	240
	Deflection @ Grade (mm)	0	0.6	2.8	6.5	11.6	18.2	28.1
	Max BM (kNm) - pile head	0	32	83	144	213	288	360
406mm (V = 550kN)	Lateral Load (kN)	0	40	80	120	160	200	240
	Deflection @ Grade (mm)	0	0.6	2.8	6.5	11.7	18.4	29.2
	Max BM (kNm) - pile head	0	32	83	144	214	289	357
406mm (V = 675kN)	Lateral Load (kN)	0	40	80	120	160	200	240
	Deflection @ Grade (mm)	0	0.7	2.9	6.6	11.7	18.7	30.6
	Max BM (kNm) - pile head	0	32	83	145	215	290	353
406mm (V = 800kN)	Lateral Load (kN)	0	40	80	120	160	200	240
	Deflection @ Grade (mm)	0	0.7	2.9	6.6	11.8	19.0	32.5
	Max BM (kNm) - pile head	0	32	83	145	215	290	348

Note: Theoretical pile wall thickness is 9.5mm. 1.5mm corrosion allowance on wall thickness considered in analysis.



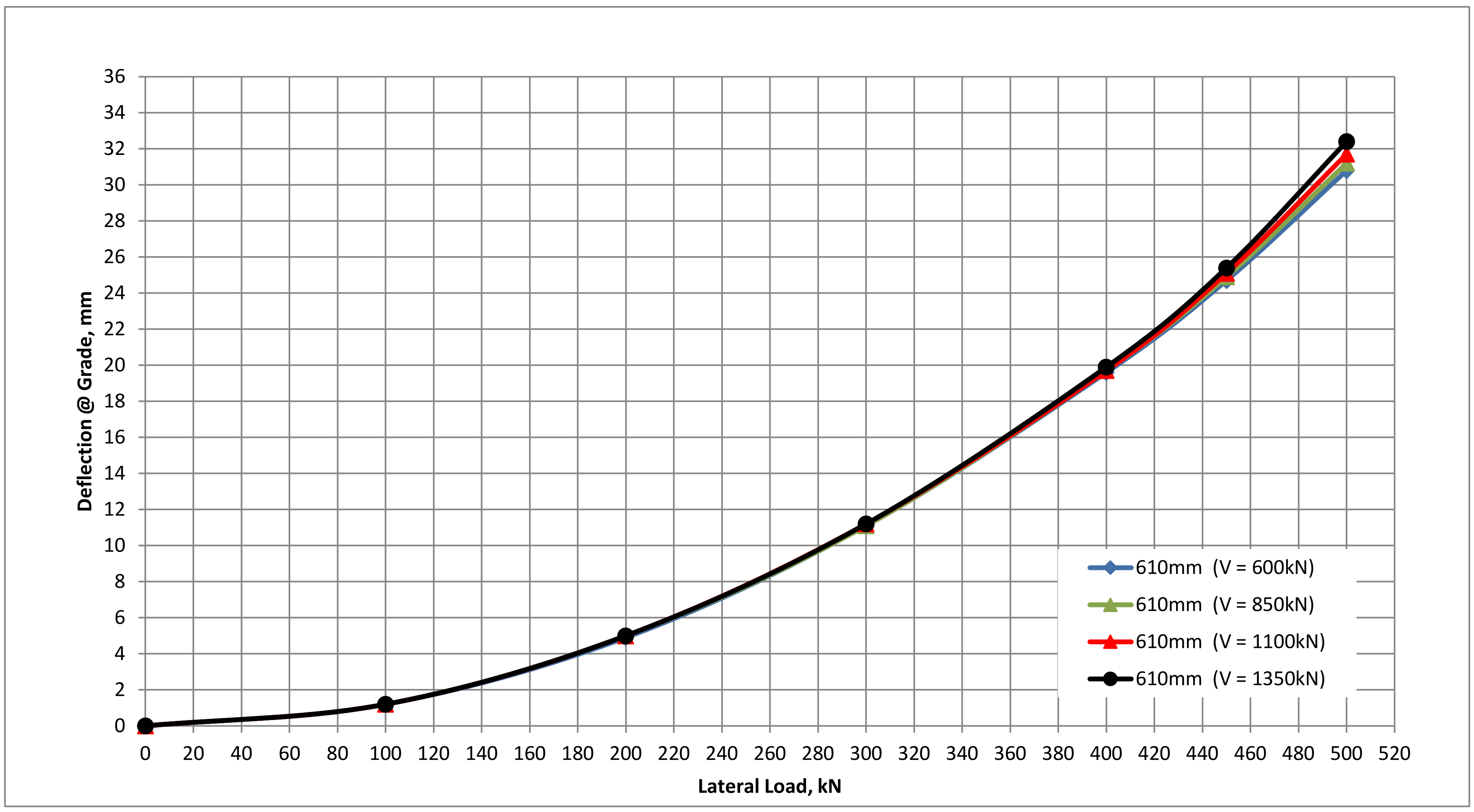


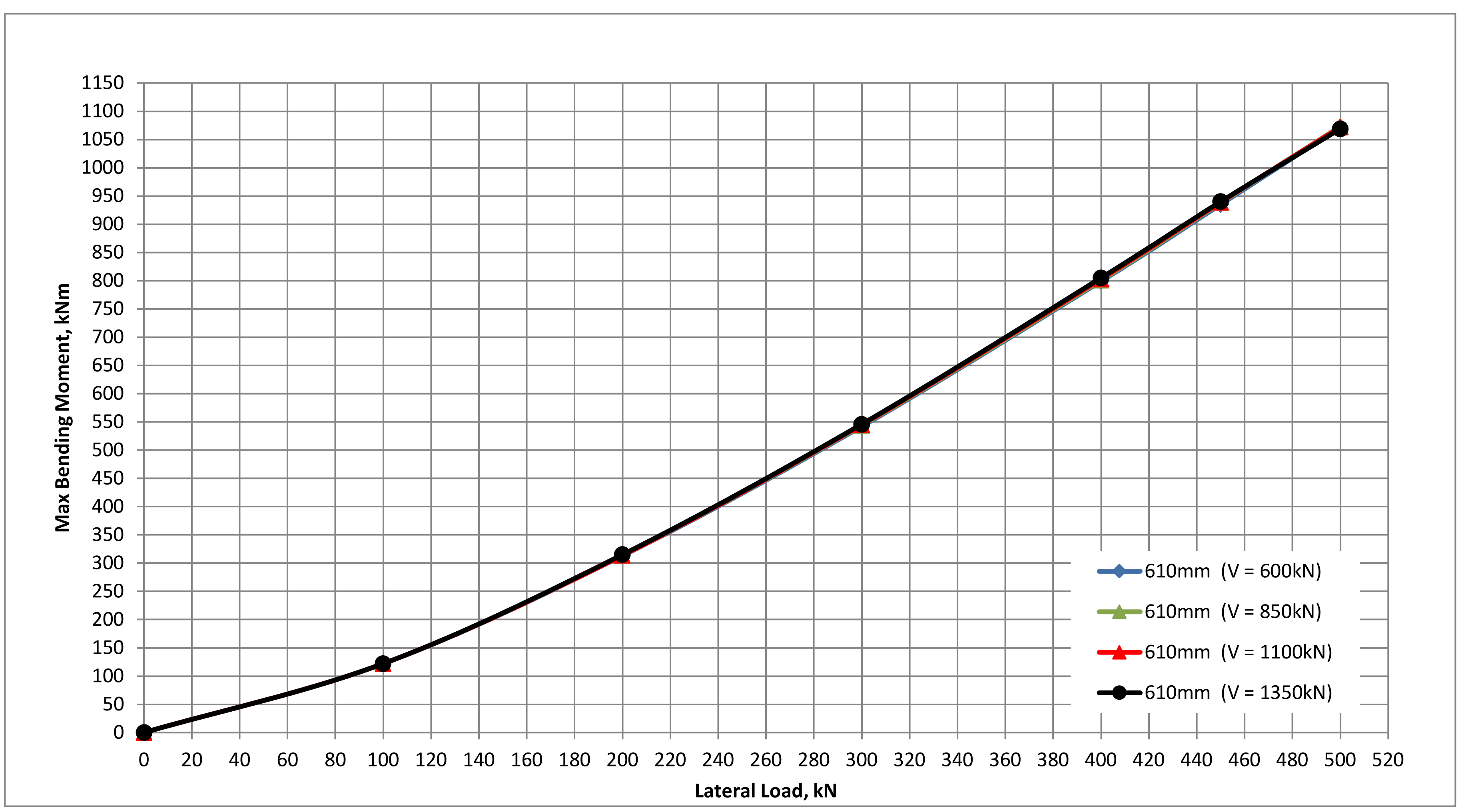
Project: SHRED
Pile Size: 610mm Dia. Pipe Pile

FIXED-HEAD STEEL PILES - PILE HEAD AT EL 2164 FT (or EL 659.5M)

610mm (V = 600kN)	Lateral Load (kN)	0	100	200	300	400	450	500
	Deflection @ Grade (mm)	0	1.2	4.9	11.1	19.6	24.7	30.8
	Max BM (kNm) - pile head	0	122	313	542	799	935	1072
610mm (V = 850kN)	Lateral Load (kN)	0	100	200	300	400	450	500
	Deflection @ Grade (mm)	0	1.2	5.0	11.1	19.7	24.9	31.2
	Max BM (kNm) - pile head	0	122	314	544	801	938	1073
610mm (V = 1100kN)	Lateral Load (kN)	0	100	200	300	400	450	500
	Deflection @ Grade (mm)	0	1.2	5.0	11.2	19.7	25.1	31.7
	Max BM (kNm) - pile head	0	122	314	545	803	939	1072
610mm (V = 1350kN)	Lateral Load (kN)	0	100	200	300	400	450	500
	Deflection @ Grade (mm)	0	1.2	5.0	11.2	19.9	25.4	32.4
	Max BM (kNm) - pile head	0	122	315	546	805	940	1069

Note: Theoretical pile wall thickness is 12.7mm. 1.5mm corrosion allowance on wall thickness considered in analysis.



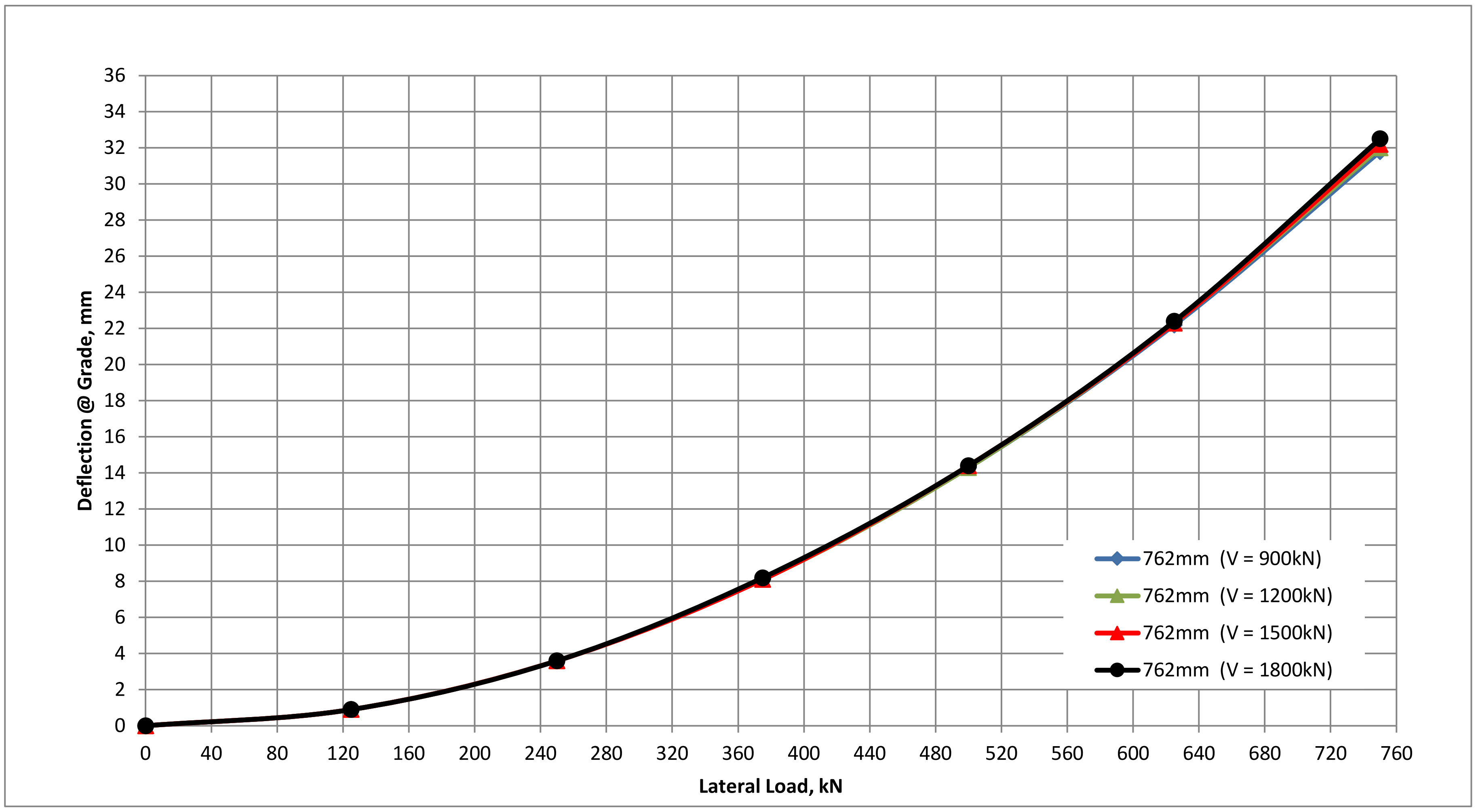


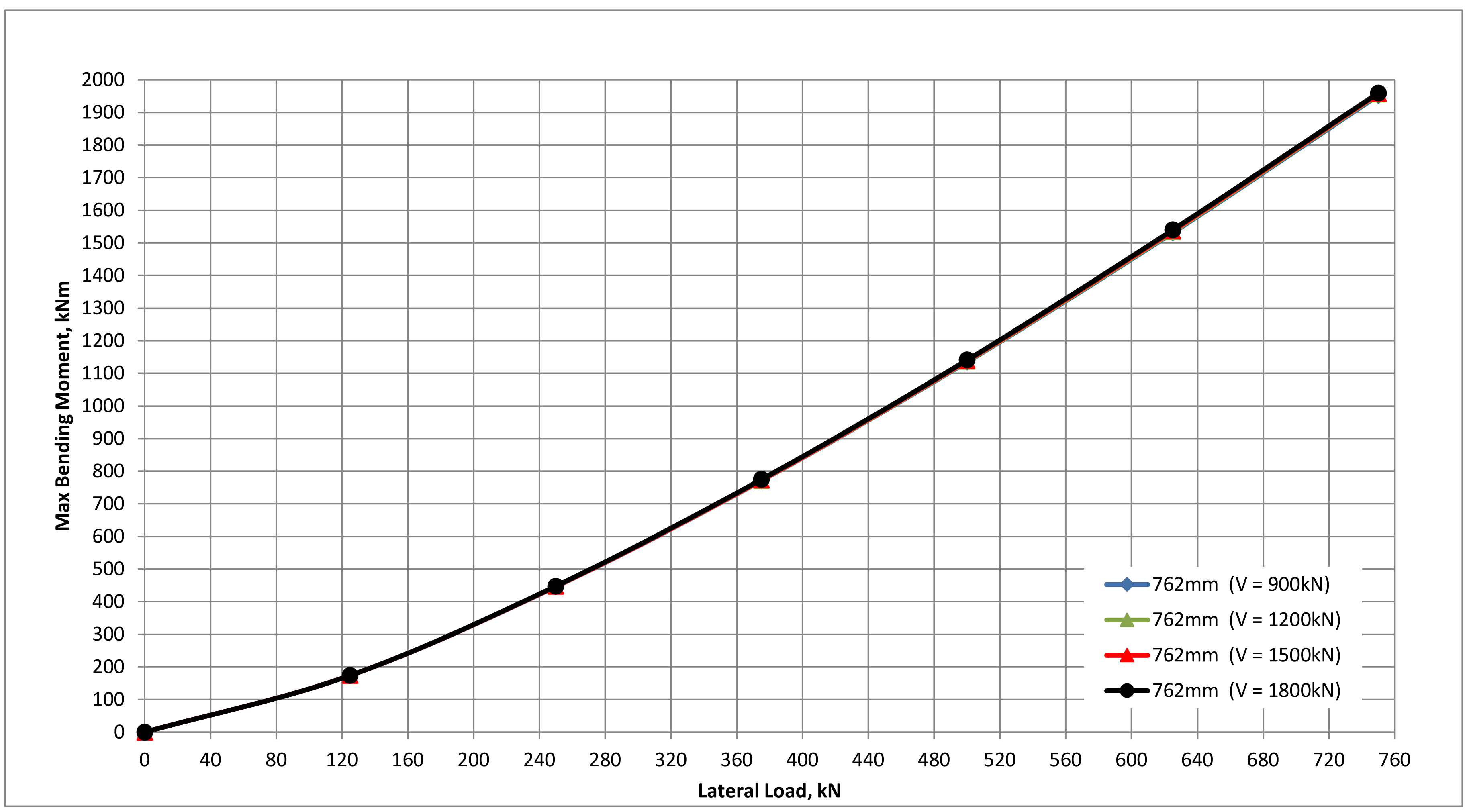
Project: SHRED
Pile Size: 762mm Dia. Pipe Pile

FIXED-HEAD STEEL PILES - PILE HEAD AT EL 2164 FT (or EL 659.5M)

762mm (V = 900kN)	Lateral Load (kN)	0	125	250	375	500	625	750
	Deflection @ Grade (mm)	0	0.9	3.6	8.1	14.3	22.2	31.8
	Max BM (kNm) - pile head	0	174	446	771	1135	1531	1952
762mm (V = 1200kN)	Lateral Load (kN)	0	125	250	375	500	625	750
	Deflection @ Grade (mm)	0	0.9	3.6	8.1	14.3	22.3	32.0
	Max BM (kNm) - pile head	0	174	447	772	1137	1534	1956
762mm (V = 1500kN)	Lateral Load (kN)	0	125	250	375	500	625	750
	Deflection @ Grade (mm)	0	0.9	3.6	8.1	14.4	22.3	32.2
	Max BM (kNm) - pile head	0	174	447	773	1139	1537	1958
762mm (V = 1800kN)	Lateral Load (kN)	0	125	250	375	500	625	750
	Deflection @ Grade (mm)	0	0.9	3.6	8.2	14.4	22.4	32.5
	Max BM (kNm) - pile head	0	174	448	775	1141	1540	1960

Note: Theoretical pile wall thickness is 15.9mm. 1.5mm corrosion allowance on wall thickness considered in analysis.





Project: SHRED
Pile Size: 914mm Dia. Pipe Pile

FIXED-HEAD STEEL PILES - PILE HEAD AT EL 2164 FT (or EL 659.5M)

914mm (V = 1200kN)	Lateral Load (kN)	0	150	300	450	600	750	900
	Deflection @ Grade (mm)	0	0.7	2.8	6.3	11.1	17.3	24.9
	Max BM (kNm) - pile head	0	232	594	1025	1508	2034	2599
914mm (V = 1500kN)	Lateral Load (kN)	0	150	300	450	600	750	900
	Deflection @ Grade (mm)	0	0.7	2.8	6.3	11.1	17.3	25.0
	Max BM (kNm) - pile head	0	232	594	1026	1510	2037	2602
914mm (V = 1800kN)	Lateral Load (kN)	0	150	300	450	600	750	900
	Deflection @ Grade (mm)	0	0.7	2.8	6.3	11.2	17.3	25.0
	Max BM (kNm) - pile head	0	232	594	1027	1511	2039	2606
914mm (V = 2100kN)	Lateral Load (kN)	0	150	300	450	600	750	900
	Deflection @ Grade (mm)	0	0.7	2.8	6.3	11.2	17.4	25.1
	Max BM (kNm) - pile head	0	232	595	1027	1513	2041	2609

Note: Theoretical pile wall thickness is 19.0mm. 1.5mm corrosion allowance on wall thickness considered in analysis.